



D^2 -RD-UNet: A dual-stage dual-class framework with connectivity correction for hepatic vessels segmentation

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ABSTRACT

Accurate segmentation of hepatic and portal veins is critical for preoperative planning in liver surgery, especially for resection and transplantation procedures. Extensive anatomical variability, pathological alterations, and inherent class imbalance between background and vascular structures challenge this task. Current state-of-the-art deep learning approaches often fail to generalize across patient variability or maintain vascular topology, thus limiting their clinical applicability. To overcome these limitations, we propose the D^2 -RD-UNet, a dual-stage, dual-class segmentation framework for hepatic and portal vessels. The D^2 -RD-UNet architecture employs dense and residual connections to improve feature propagation and segmentation accuracy. Our D^2 -RD-UNet integrates advanced data-driven preprocessing, a dual-path architecture for 3D and 4D data, with the latter concatenating computed tomography (CT) scans with four relevant vesselness filters (Sato, Frangi, OOF, and RORPO). The pipeline is completed by the first developed postprocessing multi-class vessel connectivity correction algorithm based on centerlines. Additionally, we introduce the first radius-based branching algorithm to evaluate the model's predictions locally, providing detailed insights into the accuracy of vascular reconstructions at different scales. In order to make up for the scarcity of well-annotated open datasets for hepatic vessels segmentation, we curated AIMS-HPV-385, a large, pathological, multi-class, and validated dataset on 385 CT scans. We trained different configurations of D^2 -RD-UNet and state-of-the-art models on 327 CTs of AIMS-HPV-385. Experimental results on the remaining 58 CTs of AIMS-HPV-385 and on the 20 CTs of 3D-IRCADb-01 demonstrate superior performances of the D^2 -RD-UNet variants over state-of-the-art methods, achieving robust generalization, preserving vascular continuity, and offering a reliable approach for liver vascular reconstructions.

1. Introduction

Liver cancer, particularly hepatocellular carcinoma (HCC), is a leading cause of cancer-related mortality globally. Surgical treatments, including resection and transplantation, remain the most effective options, with 5-year survival rates as high as 80% when feasible [1, 2]. Accurate preoperative planning is critical and relies on advanced imaging techniques, such as multiphase contrast-enhanced Computed Tomography (CT) or Magnetic Resonance Imaging (MRI), to assess tumor characteristics and vascular involvement, as recommended by the National Comprehensive Cancer Network® (NCCN®) Guidelines [3].

Traditional 2D imaging, though widely used, has limitations in visualizing the complex spatial relationships of hepatic structures. 3D reconstruction of hepatic vasculature provides a more comprehensive, patient-specific perspective, enabling surgeons to understand vascular topology better and improving surgical planning. Additionally, 3D visualization has been shown to improve surgical safety, reduce intraoperative risks, and lower postoperative complications [4–7].

The accurate reconstruction of hepatic (HV) and portal vessels (PV) is a fundamental challenge in medical imaging, with significant implications for preoperative planning in liver surgery, including resection and transplantation [8–11]. A precise representation of the

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liver vasculature is crucial for optimizing surgical outcomes, minimizing postoperative complications, and managing complex vascular pathologies [8,10,12]. Despite the growing use of advanced imaging techniques such as computed tomography angiography (CTA), accurate segmentation and reconstruction of the vessels still need to be improved due to the intricate and variable anatomy, combined with imaging limitations such as low contrast and motion artifacts [13,14].

1.1. Open challenges

One of the primary challenges in the liver vessels segmentation task arises from the extensive anatomical variability of hepatic and portal vessels, which differ significantly among patients [13]. Existing segmentation approaches often fail to generalize effectively across this variability, mainly when pathological conditions such as lesions and/or cirrhosis distort the vessel structures. This problem highlights the need for robust models to handle such inter-patient variability [15,16]. Furthermore, the inherent class imbalance between the background and the target structures remains a significant obstacle for current state-of-the-art (SOTA) methods. Despite progress, most segmentation models struggle to capture the smaller vascular structures accurately, often overshadowed by the predominance of background pixels [17]. Additionally, maintaining the topological continuity of the vascular network during segmentation is critical, yet current approaches frequently produce fragmented or discontinuous vessel maps. This limitation reduces the clinical reliability of these methods, as such disruptions ruin the accurate reconstruction of the vascular anatomy [18]. While recent advancements have improved accuracy on benchmark datasets, the lack of solutions that explicitly address these challenges in a “ready-to-use” framework for clinical applications remains a major gap in the field.

1.2. Public datasets and evaluation methods limitations

A clinically feasible segmentation algorithm must achieve high accuracy and strong generalization across data from different sources [19–21]. Despite the promising advancements and benchmarks in deep learning (DL)-based methods, several limitations remain, which can be summarized as follows:

- *Lack of well-annotated datasets.* Extensive evaluation of automated models is becoming increasingly essential for their adoption in clinical practice [22,23]. However, the availability of large, well-annotated datasets remains a significant challenge. This limitation hinders the development of generalizable segmentation models, as the dataset size and quality fail to capture the full spectrum of anatomical variability seen in the real-world population [24]. A literature review revealed only two public datasets for hepatic vessel segmentation: the 3D-IRCADb-01 dataset [25], and the Medical Segmentation Decathlon (MSD) dataset [26]. The 3D-IRCADb-01 labeled 20 subjects with HV and PV separately but presents limited annotations, under- and over-segmentation issues [14,24]. Meanwhile, MSD consistently reports more cases (434 subjects), but its annotations are of lower quality, resting on a single class that merges HV and PV. As our work focuses on the multiclass segmentation of HV and PV, the MSD dataset is excluded from our study evaluations.
- *Lack of suitable evaluations methods.* Evaluating anatomical structures of different sizes, such as the trunk of the inferior cava vein (ICV) and the portal vein bifurcation at the hepatic hilum, as well as smaller branches of the hepatic and portal veins, is challenging for standard metrics because the acceptable margin of error differs across these features. Therefore, it is crucial to explore additional metrics beyond the classical Dice Score Coefficient that can more accurately capture the quality of segmentation performance.

1.3. Contributions

A multi-stage, multi-class pipeline has been developed to address these open challenges, integrating advanced DL techniques with vessel-tailored preprocessing strategies and a postprocessing vessel connectivity correction algorithm. This flexible approach is rooted in a comprehensive statistical analysis of 327 subjects from our internal dataset (see Sections 3.1 and 4). This allowed us to determine an optimal set of preprocessing parameters and procedures for robust and accurate vessel segmentation. A central aspect of our approach is the implementation of a dual-pipeline system. The first pipeline processes standard computed CTA scans (from now on, the 3D approach), while the second pipeline enhances these images by concatenating them with a series of four vesselness filters (from now on the 4D approach): Sato, Frangi, Optimally Oriented Flux (OOF), and the Ranking Orientation Responses of Path Operators (RORPO) [27]. The concatenation results in a 4D hypervolume, where the CTA context is enriched with comprehensive information from the entire set of applied filters, enhancing the representation of vascular structures. The 3D and 4D approaches exploit our D^2 -RD-UNet segmentation architecture to predict the PV and HV classes, combining principles of ResNet [28] and DenseNet [29]. This architecture leverages the strengths of both networks, improving the propagation of features through the layers while maintaining the dense connections that ensure a better contextual understanding of complex vessel structures. The D^2 -RD-UNet was designed to operate in both 3D and 4D modalities, fully exploiting the advantages of our vessel-enhanced 4D hypervolumes to deliver high-fidelity reconstructions of the liver vasculature, as proposed by Garret et al. [30]. To further enhance segmentation accuracy, we developed a novel centerline-based postprocessing correction method that, to the best of our knowledge, represents the first approach to leverage vessel centerlines to refine and correct misclassifications in HV and PV segmented masks. This approach is particularly effective in restoring vessel continuity and ensuring anatomically consistent reconstructions. Our main contributions are summarized below.

1. We developed the first data-driven preprocessing pipeline. Unlike conventional SOTA methods with fixed parameters, our approach dynamically optimizes preprocessing based on anatomical variability and Hounsfield Unit (HU) distributions. This ensures robust, generalizable results, enhancing vessel contrast, topology preservation, and segmentation accuracy, providing a strong and reproducible basis for automated models.
2. Our D^2 -RD-UNet framework achieves multi-class segmentation, distinguishing hepatic and portal veins from the liver parenchyma in both 3D and 4D configurations. The 4D setup integrates CT data with vesselness filters (Sato, Frangi, OOF, RORPO), while the encoder combines residual and dense blocks to ensure feature reuse and identity propagation, preserving critical high-level vascular features. Despite being less complex than other SOTA architectures like nnU-Net, UNet-AG, VNet, SwinUNETR, our design outperforms them in accuracy and topological continuity, demonstrating its suitability for this specific task.
3. We introduced the first automatic segmentation correction algorithm based on centerlines and endpoints, improving the accuracy and reliability of vessel segmentations.
4. We developed the first radius-based local metric to address the challenges outlined in Section 1.2 regarding the lack of suitable evaluation methods. This metric offers a novel approach to analyze vascular segmentation results, focusing on the quality of individual branches retrieved according to their mean radius. This new evaluation metric allows surgeons to better assess the quality of automatic segmentation to understand where the model is more or less trustworthy. Additionally, it is a valuable tool for evaluating vessel quality in critical regions, such as lesion boundaries or anatomical landmarks, which are essential for Hepato-Biliary-Pancreatic (HPB) surgery and transplantation.

5. Our D^2 -RD-UNet framework outperformed SOTA models like nnU-Net, VNet, UNet-AG, and SwinUNETR in PV and HV segmentation on the test portion of our curated internal dataset and on 3D-IRCADb-01 dataset, with the latter being used for generalization.

2. Related works

2.1. Rule-based methods

Initial research works on liver vessel segmentation focused on traditional, non-DL techniques centered around vessel enhancement and classical image processing. These approaches proved effective even with limited data, using algorithms such as vessel context-based voting, graph cuts, and active contour models. For instance, Chi et al. [31] introduced a vessel context-based voting to identify liver vasculatures using region-based features on CT images. Similarly, Zeng et al. [32] combined anisotropic diffusion, oriented flux measures, centerline extraction, and graph cuts to address the complexities of vessel segmentation, maintaining a strong balance between noise suppression and vessel enhancement. Later, Sangsefidi et al. [33] applied a minimum-cut/maximum-flow energy minimization framework that found the global minimum cost functions and used graph-cut algorithms to segment vessel voxels accurately. Yang et al. [34] designed an automated method for extracting the liver, portal vessels, and hepatic veins from CT images. It exploited an automatic seed point identification technique, customized level set methods, and an adaptive thresholding algorithm to guarantee precise and efficient segmentation. In parallel, Chung et al. [35] proposed an active contour model (ACM) using vessel probability maps and dense vessel candidates to guide initial contour placement, enhancing vein segmentation accuracy. Later, Lebre et al. [36] developed an automatic method to extract liver vasculature for Couinaud liver classification. It extracted the centerlines using a decision-tree approach and then reconstructed the 3D venous tree through vesselness filters and a fast marching phase. Subsequently, Guo et al. [37] developed a 3D graph cut algorithm for liver vessel segmentation. The vessel centerline was extracted using a thinning algorithm, and a final step connected under-segmented vessels, ensuring a complete vascular structure. Recently, Lamy et al. [27] benchmarked and compared seven different vesselness filters across three public datasets to quantitatively and qualitatively assess their performance, contributing valuable insights into vessel enhancement techniques. These methods laid the groundwork for DL models by providing robust preprocessing techniques to enhance liver vasculature.

2.2. Deep learning-based methods

The advent of DL has reshaped hepatic vessel segmentation, offering new and improved approaches to address its challenges. Huang et al. [17] proposed a 3D UNet with data augmentation and a Dice-based loss function to segment liver vessels, addressing class imbalance. Kitrungrotsakul et al. [38] proposed a multi-pathway DL network for liver vessel segmentation. This method uses three 2D orthogonal patches (sagittal, coronal, and axial planes) for patch-based classification, concatenating features from each volume plane. Golla et al. [39] applied a combination of CNN-based architectures, including UNet, VNet, and DeepVesselNet, trained on the 3D-IRCADb-01 dataset to automatically extract arterial and venous vessels. Survarachakan et al. [40] employed a 3D UNet architecture with $64 \times 64 \times 64$ CTA patches and explored different preprocessing and postprocessing techniques based on different vesselness filter combinations. Li et al. [41] introduced a hierarchical deep attention network that used semi-supervised training with uncertainty awareness, offering significant improvements in robustness when dealing with noisy data. Affane et al. [42] introduced an end-to-end deep learning method for vessel segmentation using vessel enhancement filters within a 3D U-Net-based architecture. 3D

U-Net, DenseU-Net, and MultiResU-Net models were compared on the 3D-IRCADb-01 dataset, with vesselness filters improving segmentation, especially at bifurcations. Xu et al. [43] took a new approach with the Mean-Teacher Confident Learning framework, leveraging high-quality and noisy data for robust segmentation. Svobodova et al. [14] tested a 3D-UNet and a 3D VNet with different vesselness filter combinations, showing how vesselness filter integration improves the segmentation results. Later, Wu et al. [44] developed IBIMHAV-Net, a 3D Swin Transformer-based network that integrated convolution and self-attention, leading to precise voxel-wise embedding and robust liver vessel segmentation. Alirri and Rahni [45] designed a ResDense UNet 2D network that merged ResNet and DenseNet architectures with advanced preprocessing steps like anisotropic diffusion. Tong et al. [46] proposed a lightweight contextual and morphological awareness network incorporating a 3D reconstruction module for improved vessel continuity and structure. Li et al. [47] introduced LU-Net, a 3D network with a graphical connectivity constraint module (GCCM) at the end to restore connectivity. Zhou et al. [48] proposed a 2D UNet-based architecture with a graphical module in the bottleneck, enhancing feature extraction through attention-based skip connections. Xu et al. [49] introduced a dual-stream encoder network combining convolution and transformer blocks for improved segmentation and employing 3D CTA patches of shape $96 \times 96 \times 96$. Chen et al. [50] developed a 3D shape-guided model using limited 2D vessel annotations. Chen et al. [51] presented TransUNet, a framework integrating Transformer-based global context extraction with CNN feature maps. Li et al. [52] proposed a coarse-to-fine 3D deep learning algorithm, starting with coarse segmentation for general feature extraction, followed by fine segmentation to refine details, effectively handling anatomical variations in the hepatic region. Garret et al. [30] also explored the benefits of using vesselness filters for a UNet model. Their work used six vesselness filters as additional modalities in a 4D hypervolume and exploited them in early fusion. Lastly, Xu et al. [53] introduced VSNet, a multi-task model with a vessel-growing decoder designed to enhance connectivity and topological consistency in minor hepatic and portal veins.

3. Datasets

In this section, we present the data used in this study. First, we describe the internal dataset AIMS-HPV-385, specifically segmented to meet clinical requirements and exploited to design a robust preprocessing pipeline. Subsequently, we introduce the publicly available dataset 3D-IRCADb-01 used to test the generalizability of the models, highlighting its limitations and challenges. The main features of the two datasets are provided in Table 1.

3.1. AIMS-HPV-385 dataset

As previously mentioned, the topology of the hepatic vasculature is highly complex, with significant anatomical variability. Moreover, the level of detail required for clinical applicability extends beyond the main vessels alone. Therefore, accurately segmenting the HV and PV requires a well-annotated dataset with sufficient size and diversity to capture the full statistical context of the task, including anatomical variability. Given these considerations, the Fondazione MIAS (AIMS Academy) in Milan (Italy) gathered and annotated a proper dataset to address these challenges.

The AIMS-385 dataset was composed of 385 abdominal CTA scans collected between January and December 2021, consisting of anonymized DICOM images. These CTAs followed a standardized protocol, starting with a non-contrast phase and then injecting an iodinated contrast agent. The arterial phase (AP) was captured 20–30 s after injection, the portal venous phase (PVP) at 60–70 s, and a delayed phase at 3–4 min, highlighting contrast retention. This protocol effectively captured the liver parenchyma (LV) and both arterial and venous vasculature [54].

Table 1
Properties of AIMS-HPV-385 and 3D-IRCAdb-01 datasets.

Name	Country	Size	Images	Phase	Labels	Resolution [h × w × d] Spacing [mm]	Annotation	Pathological
AIMS-HPV-385	Italy	385	CTA	Portal Venous	Hepatic Vessels Portal Vessels Liver Parenchyma	512 × 512 × [69:859] [0.59 : 0.98] × [0.59 : 0.98] × [0.62 : 3.00]	Manual	✓
3D-IRCAdb-01	France	20	CTA	Portal Venous	Hepatic Vessels Portal Vessels Liver Parenchyma Liver Tumor	512 × 512 × [74:260] [0.56 : 0.87] × [0.56 : 0.87] × [1.00 : 4.00]	Manual	✓

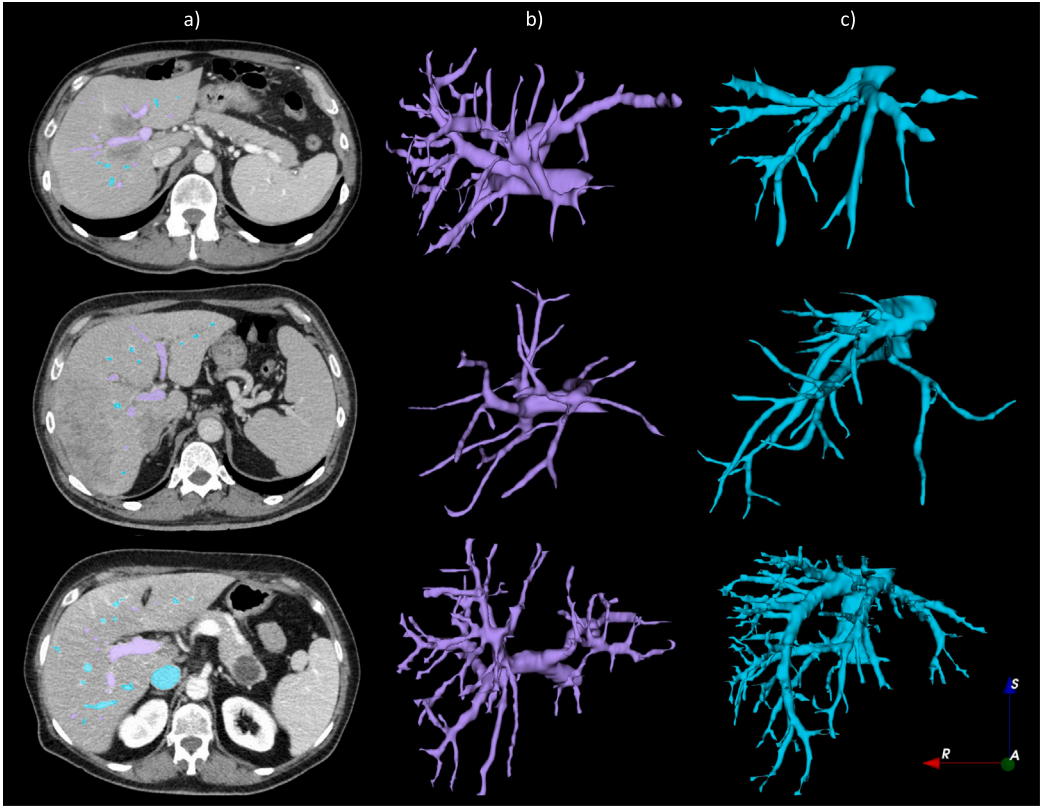


Fig. 1. Three samples of AIMS-385 dataset: 2D axial view of CTAs in PVP (a), and 3D perspective view of PV (b) and HV (c) ground truth.

After data collection, the AIMS-385 dataset was annotated to label the HV, PV, and LV on the portal venous phase of the CTA scans. Each volume had an axial resolution of 512×512 pixels, with slice counts ranging from 69 to 859. The spacing parameters ($h \times w \times d$) varied: h and w ranged between 0.59 mm and 0.98 mm, while d ranged from 0.62 mm to 3.00 mm.

To ensure accuracy and minimize human error, a team of five clinical annotators, each with at least one year of experience, received specialized training from expert radiologists with extensive expertise in the field. The manual annotation process was conducted using 3D Slicer.² Given the dataset's complexity and vein morphology variability, a panel of expert radiologists and senior physicians developed and provided detailed annotation guidelines. These guidelines emphasized the importance of accurately capturing complex vascular branches, avoiding discontinuities unrelated to tumor interference, and thoroughly verifying potential structural contacts to ensure high-quality annotations.

The expert physicians conducted the review process in two stages to ensure inter-expert consensus on the dataset: during the annotation phase and after completing the initial annotations. Uncertain

labels were addressed in consensus meetings, where the experts jointly reviewed problematic annotations and selected the most accurate representation. Additionally, random spot checks were performed on 60% of the annotated images to assess label accuracy and compliance with the guidelines. The annotators corrected any anomalies detected during these two review stages. Fig. 1 illustrates three distinct cases from the AIMS-385 CTA dataset, along with their corresponding PV and HV ground-truth (GT) reconstructions.

3.2. 3D-IRCAdb-01 dataset

The 3D-IRCAdb-01 dataset includes 20 contrast-enhanced CT scans, 75% featuring hepatic tumors. Although this dataset was frequently employed in advanced liver vessel segmentation studies, it raised several concerns regarding its broader applicability and limitations. For instance, some CT volumes exhibited incomplete annotations or other issues that could impact segmentation performance. Despite their clear visibility in the images, the hepatic HV needed to be more adequately annotated. The liver sometimes did not include the inferior vena cava (IVC) [13,14]. These limitations reduced the exploitability of this dataset in developing clinically applicable solutions, unmeeting the level of detail and precision required by the surgeons. Furthermore,

² www.slicer.org

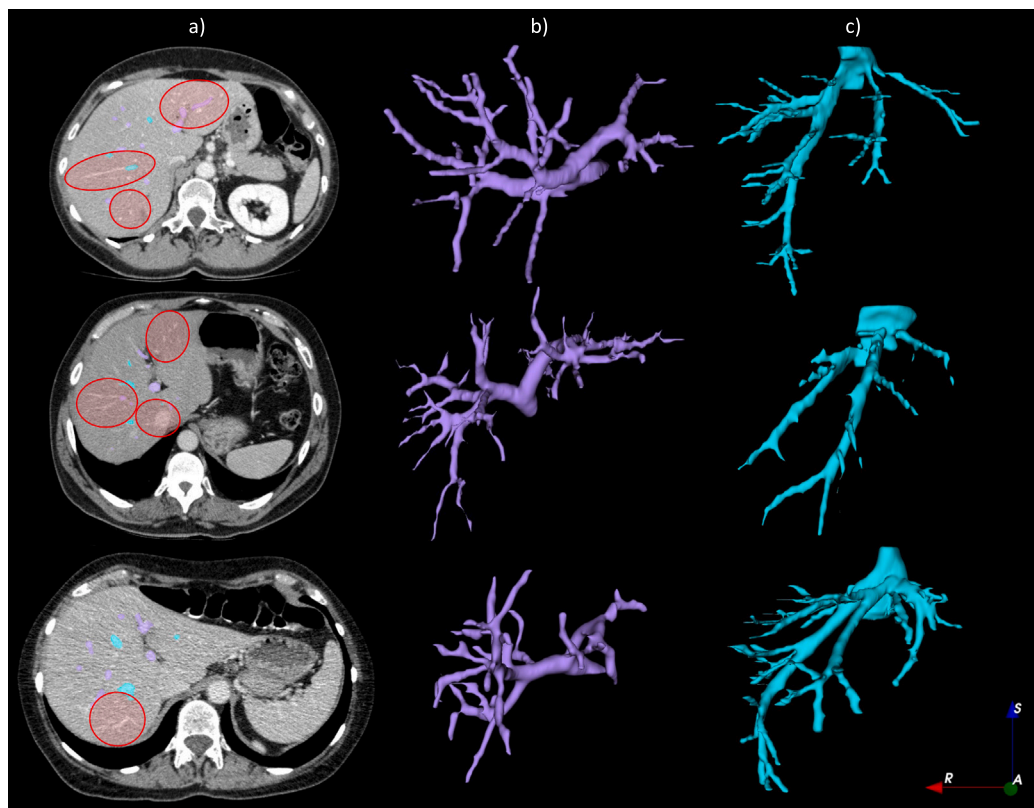


Fig. 2. Three samples of 3D-IRCADb-01 dataset: 2D axial view of CTAs in PVP (a), and 3D perspective view of PV (b) and HV (c) ground truth. From top to bottom: IR_01, IR_05, IR_15. Some major and peripheral missing annotations are circled in red.

due to its limited size, the 3D-IRCADb-01 dataset is better suited for traditional machine learning approaches, such as support vector machines, random forests, or decision trees, which typically require less data than DL models. The small dataset size would likely result in overfitting for 3D DL models, making traditional machine learning methods more appropriate for the 3D-IRCADb-01 dataset [24]. Fig. 2 shows three different cases from the 3D-IRCADb-01 dataset, along with their corresponding GT reconstructions of the PV and HV.

To better quantify the limitations of the 3D-IRCADb-01 dataset and support the rationale behind our evaluation protocol, we conducted a detailed inspection of all annotated volumes in collaboration with an expert radiologist. This analysis focused on two key sources of inconsistency:

- Peripheral vessel truncation, where annotations do not extend to the finer, distal branches of the vascular tree, resulting in incomplete ground truth data. We assessed whether the peripheral coverage was sufficient for each volume to represent the whole hepatic vasculature well. The red circles in Fig. 2 highlight this lack.
- Inclusion of extrahepatic vascular structures, which extends beyond the intended scope of hepatic vessel segmentation. Precisely, for HV, we considered the suprarenal and infrarenal portions of the IVC as out-of-scope. For PV, we identified as extrahepatic the inclusion of the splenic vein (SV), superior mesenteric vein (SMV), inferior mesenteric vein (IMV), and gastric veins (GV). These structures are anatomically valid but lie outside the hepatic domain and may introduce several issues: (a) they increase the apparent vascular volume, potentially biasing segmentation metrics; (b) they distort topological coherence within the liver parenchyma; and (c) their partial and inconsistent annotation across cases may lead to misleading model evaluations. The Fig. 3 shows these out-of-scope regions as semi-transparent areas.

As thoroughly reported in Table A.1, only three out of 20 HV and four out of 20 PV segmentations provided adequate peripheral detail, and 14 out of 20 HV and 15 out of 20 PV segmentations included out-of-scope vascular extensions. Because of this high percentage of inaccurate annotations, we decide to exclude 3D-IRCADb-01 from model training and to use it exclusively as a technical benchmark. Furthermore, to ensure reliable and anatomically focused evaluations, our testing protocol included only the vascular structures located within the hepatic region, explicitly removing all extrahepatic segments prior to evaluation. The cleaned version of the dataset, which includes volumes like those shown in Fig. 3, allowed for a more accurate and clinically relevant assessment of model performance. Despite its limitations, the 3D-IRCADb-01 dataset still remains the most publicly used benchmark due to its inclusion of complex pathological cases, such as large tumors and imaging artifacts (e.g., metallic prostheses), which are rarely found in other publicly available datasets. However, we emphasize that meaningful use of this dataset as a clinical benchmark would require a thorough expert-driven re-annotation process.

4. Methods

In this section, we describe the detailed methods used in our study to develop a robust and accurate pipeline for liver vessel segmentation from CTA scans. The methods are divided into four main stages: preprocessing, vesselness filters, network design, and postprocessing. Each stage was designed to address the unique challenges posed by the complex anatomical and imaging characteristics of hepatic vasculature, ensuring the integrity and accuracy of vessel structures throughout the segmentation process. Table 2 provides a schematic overview, while an overall pseudocode of our methods is available at the end of the section in Algorithm 1 for further clarity. At the end of the manuscript, Table 9 reports a list of abbreviations and symbols.

Table 2
Schematic overview of the methods presentation.

	Methods outline
Section 4.1: Preprocessing	Section 4.1.1: Data preparation Section 4.1.2: Data-driven HU windowing
Section 4.2: Vesselness filters	Section 4.2.1: Hessian-based methods Section 4.2.2: Flux-based methods Section 4.2.3: Morphology-based methods Section 4.2.4: Filters parameters tuning
Section 4.3: DL Network	Section 4.3.1: Dual input pipeline Section 4.3.2: RD-encoder Section 4.3.3: RD-decoder Section 4.3.4: Output reconstruction Section 4.3.5: Loss function Section 4.3.6: Global evaluation metrics Section 4.3.7: Local radius-based metrics
Section 4.4: Postprocessing	Section 4.4.1: Vessels connectivity correction

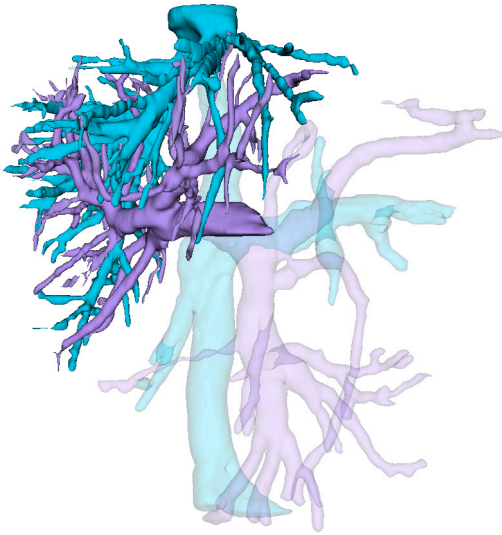


Fig. 3. Example of the 3D visualization of the hepatic and portal venous systems of one subject (ID: IR_01) of 3D-IRCADB-01 dataset. Transparent volumes indicate vessel segments that extend beyond the liver-specific vascular territory.

The 3D volumes employed in the 3D pipeline were considered as $x \in \mathbb{R}^{H \times W \times D \times C}$, with channels $C = 1$ corresponding to grayscale CTA images. For the 4D volumes, the 4D pipeline was considered as $x \in \mathbb{R}^{H \times W \times D \times C}$, with $C = 5$, including the grayscale CTA image and the four vesselness filter outputs. During the training phase, both 3D and 4D volumes were extended by incorporating the batch size dimension B , resulting in representations $x \in \mathbb{R}^{B \times H \times W \times D \times C}$. This dimensionality accounted for the efficient processing of multiple volumes in parallel during model training.

The AIMS-HPV-385 dataset used for this study was split into a training set (TrS), consisting of 327 subjects (85%), and a test set (TeS) with the remaining 58 subjects (15%). The 20 subjects of 3D-IRCADB-01 were used as a held-out dataset for evaluation purposes.

4.1. Preprocessing

4.1.1. Data preparation

In the following subsection, the data preparation process is described. It consists of resampling, masking, and patching. Each step was thoughtfully conceived to ensure the quality and representativeness of the resulting dataset, reflecting insights drawn from an in-depth analysis of its particular characteristics.

In SOTA methods, it is typical to perform **resampling** to uniform the spatial resolution of images and their corresponding segmentations

to a common spacing [17,45]. However, we found that this approach can introduce significant biases and remove critical topological information. As shown in Fig. 4, scaling an image from 1 mm to 3 mm spacing results in the distortion of thin and horizontal structures, introducing holes and discontinuities. This distorts the segmentation masks, violating the topological continuity principles outlined in our annotation guidelines (see Section 3.1). To avoid these issues, we chose to avoid resampling. Instead, we preserved the dataset's dimensional heterogeneity, maintaining the original topological features essential for the robustness of our DL models.

We implemented a **patching mechanism** to address the challenges posed by different resolutions and slice thicknesses across the dataset [44,45]. Anisotropic patches produced better results from our analysis of SOTA techniques [55]. After testing different patch sizes, we selected $192 \times 192 \times 64$ as the optimal dimension, maintaining a short volume and a broad spatial context. This approach enables efficient dataset processing while preserving critical information; the algorithm was designed to handle both 3D volumes and 4D hypervolumes.

Furthermore, another commonly adopted approach in SOTA methods involves **liver-masking** the CTA to isolate the hepatic region. This strategy reduces interference from surrounding structures and focuses the network's attention only on the liver volume. However, this approach removes two critical topological features: the insertion of the portal vein at the hepatic hilum and the convergence of the hepatic veins into the inferior vena cava. Although this method increases the network's focus, it removes crucial information about the main originating vessels, thereby increasing uncertainty. In our two-stage pipeline, we decided not to use masking. In the first stage, we trained a 2.5D UNet on 600 subjects with manually annotated livers to localize the liver within the full CTA volume. This step allows us to identify and extract a liver-encompassing region from the original volume, reducing data dimensions, lowering computational cost, and focusing on the hepatic area. In the second stage, the actual segmentation was performed, preserving the topological information of the main vessels and ensuring high accuracy in the detailed segmentation of vascular structures.

4.1.2. Data-driven HU windowing

SOTA methods typically use a fixed HU range of [80:220] for vessel enhancement [40]. However, this range often proves too restrictive for multi-class segmentation, especially when preserving thin anatomical features across different vascular structures. To address this issue, we leveraged our large-scale AIMS-HPV-385 dataset to conduct a detailed statistical analysis of HU distributions, specifically within the HV and PV. Our goal was to define a more inclusive HU window that preserves anatomical detail more accurately. We computed the 5th and 95th percentiles of the HU distributions for each case and vascular structure to avoid outliers; an example is depicted in Figure B.1a. This analysis captured the typical HU ranges of these vessels while accounting for inter-patient and structural variability. Subsequently, we retrieved the overall distribution by aggregating these percentiles across all samples in our dataset, as shown in Figure B.1b. We then defined the new HU window by selecting the 5th and 95th percentiles from the aggregated distribution across HV and PV. This approach ensured that the final HU window captured the full range of relevant intensities, from the lowest minimum to the highest maximum, preserving fine grayscale details. The new data-driven HU range was set to [55:275] and applied for both 3D and 4D pipeline CTA volumes. In addition to the updated HU windowing strategy, we applied a Gaussian smoothing filter with a kernel size of 3 and a sigma of 2.

Our data-driven method offered two main advantages. First, it adapted to inter-patient variability and anatomical differences, ensuring subtle vascular features are retained across different subjects. Second, it mitigated the risk of losing valuable intensity details, particularly those relevant to smaller vessel branches that a narrower HU window might otherwise exclude. Expanding the HU window and applying Gaussian smoothing allowed us to enhance vessel contrast while preserving critical features, ultimately improving segmentation quality and downstream task performance.

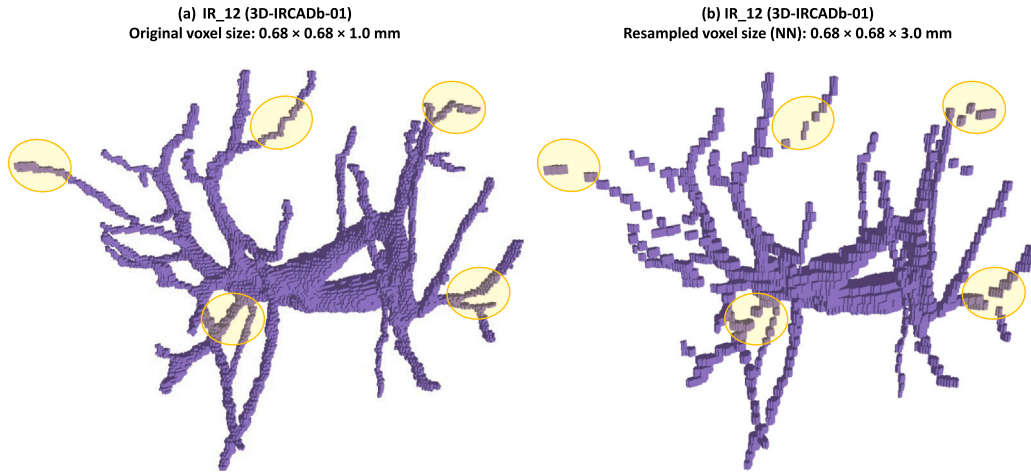


Fig. 4. Example of holes and discontinuities caused by resampling. (a): PV ground truth segmentation for subject IR_12 (3D-IRCADb-01) with original voxel size of $[0.68 \times 0.68 \times 1.0]$ mm. (b): PV ground truth segmentation for subject IR_12 (3D-IRCADb-01) with distortions introduced after resampling the spacing to 3.0 mm using nearest neighbor interpolation (NN).

4.2. Vesselness filters

Vesselness filters play a crucial role in vessel segmentation by enhancing the visibility of vascular structures in CTA scans [30,37,40,56]. These filters improve contrast in vessel-like regions while suppressing non-vascular structures and reducing noise. However, vessel enhancement remains challenging due to the high variability in vascular anatomy, both between and within patients. Vessel diameter, curvature, length, and topology can differ significantly across anatomical regions. To address this variability, vesselness filters are specifically designed to detect tubular structures representing a consistent geometric pattern in blood vessels.

In this section, we present the mathematical formulation of the four vesselness filters used in our pipeline for generating 4D hypervolumes. We also explain their relevance for enhancing tubular structures and motivate our choice based on their ability to capture vessels across a wide range of scales and geometries. We begin by presenting the three Hessian-based filters we used: Sato filter, Frangi filter, and Optimally Oriented Flux (OOF) filter. Next, we cover morphological methods, with a reminder of basic morphological operations and the presentation of the RORPO filter. Finally, we present the approach used to select the parameters for these filters.

4.2.1. Hessian-based methods

Many vesselness filters rely on the Hessian matrix to identify tubular structures by analyzing the local curvature of the intensity of voxels in the image. Since vessels exhibit a wide range of diameters, these methods typically operate in a multi-scale framework to detect both small and large vascular features. We provide a detailed explanation of the Hessian matrix, its eigenvalue interpretation, and its multi-scale formulation in Appendix C. Our work employed three well-established Hessian-based filters: Sato, Frangi, and Optimally Oriented Flux (OOF). Each filter applies a different strategy to interpret the Hessian eigenvalues and enhance curvilinear structures.

Sato filter, introduced early by Sato et al. [57], was developed for curvilinear structures in medical images. In the method, eigenvalues are sorted such as $\lambda_1 \geq \lambda_2 \geq \lambda_3$, which allows tubular structures to be conditioned at points where $\lambda_1 \approx 0$ and $\lambda_2 \approx \lambda_3 \ll 0$. In this case, e_1 , the eigenvector corresponding to λ_1 , indicates the direction of the tube while e_2 and e_3 correspond to the main directions of the tube cross-section. The vesselness function is then defined as Eq. (1):

$$F_{Sato} = \begin{cases} \lambda_c \exp\left(-\frac{\lambda_1^2}{2(\alpha_1 \lambda_c)^2}\right) & \text{if } \lambda_1 \leq 0, \lambda_c \neq 0, \\ \lambda_c \exp\left(-\frac{\lambda_1^2}{2(\alpha_2 \lambda_c)^2}\right) & \text{if } \lambda_1 > 0, \lambda_c \neq 0, \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

where $\lambda_c = \min(-\lambda_2, -\lambda_3)$ and the couple α_1, α_2 are weights that parametrize the response of the filter to tubular structures.

Frangi filter, in contrast to the method proposed by Sato et al. which utilizes only two eigenvalues, emphasizes the importance of considering all three eigenvalues, as highlighted in [58]. This filter defines three measures, as shown in Eq. (2), which are used to distinguish between different types of structures: two geometric ratios, R_b and R_a to discriminate blob-like and line-like structures from planar objects, respectively; and another ratio, S to reduce background voxel response and improve robustness to noise.

$$\begin{aligned} R_b &= \frac{|\lambda_1|}{\sqrt{|\lambda_2 \lambda_3|}} \\ R_a &= \frac{|\lambda_2|}{|\lambda_3|} \\ S &= \|H\| = \sqrt{\lambda_1^2 + \lambda_2^2 + \lambda_3^2} \end{aligned} \quad (2)$$

The vesselness function is then the combination of these three measures given in Eq. (3), where α, β , and C are weights that adjust the sensibility to tubular and line-like structures, blob-like structures, and high-variance texture respectively.

$$F_{Frangi} = \begin{cases} \left(1 - \exp\left(-\frac{R_b^2}{2\alpha^2}\right)\right) \exp\left(-\frac{R_a^2}{2\beta^2}\right) \left(1 - \exp\left(-\frac{S^2}{2C^2}\right)\right) & \text{if } \lambda_2, \lambda_3 \leq 0 \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

4.2.2. Flux-based methods

Optimally Oriented Flux (OOF) is not strictly a Hessian-based method. However, the authors demonstrated that the oriented flux matrix Q is equivalent to $Hess$ in Equation C.1 [59]. Therefore, we approached it like any other Hessian method. OOF identifies the direction of tubular structures by minimizing the inward-oriented flux, i.e., the projection of the image gradient along a specific direction at the surface of an enclosed local region. A sphere is used as a projection surface for curvilinear structures like vessels. This alternative approach has the advantage of setting its scale space from the radius of this sphere,

Table 3

Summary of the vesselness filters used, their types, and selection rationale based on structural enhancement capabilities.

Filter	Type	Motivation for selection
Sato	Hessian-based	Enhances fine tubular structures via eigenvalue analysis.
Frangi	Hessian-based	Robust to noise; distinguishes vessels from blobs and planes.
OOF	Flux-based	Uses spherical scale space; handles adjacent structures well.
RORPO	Morphology-based	Detects curvilinear paths across multiple orientations.

making the method robust to interferences of closely located adjacent objects with no additional computational load. OOF was designed as a generic curvilinear structure detector, using the geometric mean of the first two eigenvalues of Q to make it more suitable for vessel enhancement. The response for a voxel at a specific scale is described in Eq. (4).

$$F_{OOF} = \begin{cases} \sqrt{|\lambda_1 \lambda_2|} & \text{if } \lambda_1 \leq \lambda_2 < 0 \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

4.2.3. Morphology-based methods

Morphological methods offer an alternative to Hessian-based approaches by analyzing image structures through shape-driven operations. These techniques are particularly effective for detecting curvilinear patterns using directional path-based transformations. A detailed explanation of the core morphological concepts used in our work is provided in Appendix C.

RORPO filter builds on foundational work by Heijmans et al. [60], introducing the concepts of path openings (see Equation C.2) and closings (see Equation C.3). RORPO leverages a set of structuring elements (SE) that represent a variety of constrained paths. In their study, the RORPO authors applied path openings across different orientations, ranking these paths to retain features present in at least three orientations [61]. This method effectively distinguishes tubular structures from blob-like and planar formations.

4.2.4. Filters parameters tuning

As previously stated, the methods described above involve parameters that control the response to various aspects of vessel geometry. To ensure optimal filter enhancement, these parameters must be fine-tuned depending on the specific purpose and data. To achieve this, Lamy et al. [62] provided a benchmark where filtered images were computed using a set of predefined parameter combinations. The outcome of this benchmark enabled the selection of the optimal parameter combination for specific goals, e.g., enhancing vessels within a given size range. In our work, we did not focus on this optimization aspect, and we directly used the optimized parameters evaluated from 3D-IRCADb-01 presented in [62].

Table 3 summarizes the vesselness filters integrated in our pipeline, along with their classification and the rationale behind their selection. These filters collectively enable robust multi-scale vessel enhancement across diverse vascular morphologies, forming the foundation for the subsequent stages of our segmentation 4D pipeline.

4.3. Network architecture

The proposed D^2 -RD-UNet framework is a hybrid architecture designed for the specific needs of 3D liver vessel image segmentation, integrating ResNet and DenseNet features. As previously described, it was developed to handle both 3D volumes and 4D hypervolumes. Its architecture is depicted in Fig. 5. The input was formally defined as: $X \in \mathbb{R}^{B \times X \times Y \times Z \times C}$. In the 4D case, the channel C included the preprocessed CTA data and the vessels-enhanced features, providing the network with additional context for accurate segmentation. A batch size of $B = 4$ was selected.

4.3.1. Dual input pipeline

The proposed framework processes raw 3D CTA volumes through

two distinct and independent input pipelines: the 3D and 4D approaches. Both pipelines start with a liver localization step, where a custom 2.5D UNet identifies and extracts a liver-encompassing sub-volume from each full CTA scan. The data-driven HU windowing is then applied to the liver subvolume. Fig. 5 (top-right quadrant) visually represents this process (localization stage). From this stage onward, all processing steps operate exclusively on the extracted liver-encompassing regions (liver CTA). In the **3D pipeline**, the algorithm subdivides the windowed subvolume into fixed-size patches, resulting in input tensors of shape ($B = 4, 192, 192, 64, C = 1$). These patches are the inputs of D^2 -RD-UNet architecture. In the **4D pipeline**, the algorithm firstly computes four vesselness filters (Sato, Frangi, RORPO, and OOF) following the mathematical formulations detailed in Section 4.2 from each windowed subvolume. Each filter generates a 3D volume aligned with the original liver-encompassing region. The model concatenates these four filtered volumes with the windowed CTA, forming a five-channel representation. After subdividing this multi-channel volume into patches, it receives input tensors of shape ($B = 4, 192, 192, 64, C = 5$), which it feeds into the extended architecture, D^2 -RD-UNet (4D). This dual-pipeline design enables a direct and fair comparison between a standard intensity-based (3D) and enhanced multi-channel input (4D). Fig. 5 (top-left quadrant) provides a visual representation of this dual-pipeline design.

4.3.2. Encoder of D^2 -RD-UNet

The ResDense block, which combines identity mapping from ResNet [28] and feature reuse from DenseNet [29] represents the core of the D^2 -RD-UNet framework. Each block consists of two consecutive 3D convolutions, each followed by batch normalization (BN) and ReLU activation function. Instead of summing the input and output, D^2 -RD-UNet avoids feature dilution by concatenating the original input x with the output of the second convolution. This concatenation ensures that low-level and high-level features are preserved and passed forward, especially when processing reduced-resolution patches. By reusing features across layers, D^2 -RD-UNet prevents information loss, making it well-suited for segmenting small structures like hepatic and portal veins. In the 4D approach, the first convolutional layer of D^2 -RD-UNet (4D) processes the 4D hypervolume, exploiting early fusion to merge initially all the filters. During training, the network combines them optimally [30]. This early fusion mechanism provides the network with a diversified set of features from the beginning, allowing the encoder to select and integrate the most informative representations during training autonomously. By doing so, the model is not limited to a single filtering configuration but can dynamically exploit the most relevant information to optimize the loss function.

4.3.3. Decoder of D^2 -RD-UNet

The decoder path mirrors the encoder using 3D transposed convolutions to upsample feature maps and restore the original spatial resolution. Skip connections concatenate the corresponding feature maps from the encoder, ensuring that fine details and contextual information are retained, improving segmentation accuracy. Additionally, a second concatenation after the ConvBNReLU blocks in the decoder further enriches the feature space, helping to maintain detailed structure information. The network output is a multiclass segmentation map with four classes: background, HV, PV, and LV. The introduction of liver class helped the network to increase contextual understanding, reducing small isolated outliers belonging to the parenchyma.

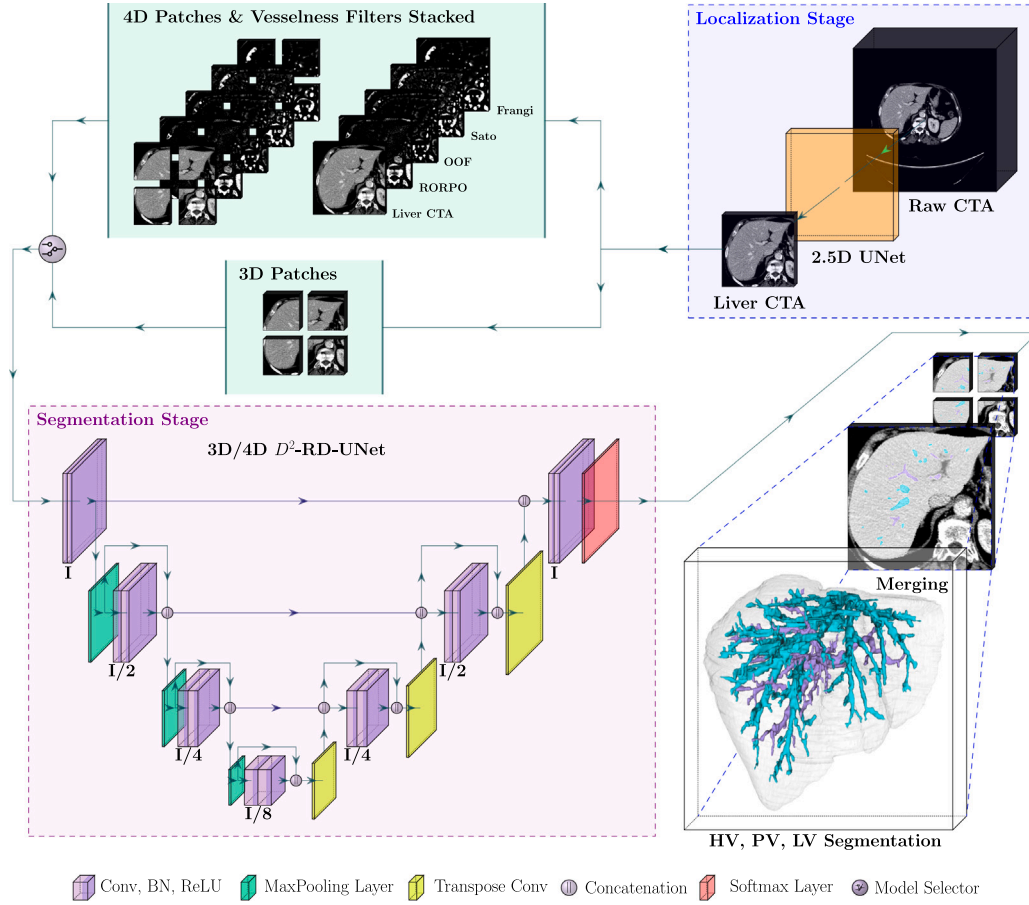


Fig. 5. Overview of D^2 -RD-UNet architecture. The dual-stage dual-class framework begins with the **localization stage**, where an initial coarse network identifies the liver and crops the CTA to a liver-encompassing region. Next, either a 3D or 4D vesselness filter-based configuration is applied with patching. The **segmentation stage** endows D^2 -RD-UNet with residual and dense blocks for multiclass HV and PV segmentation. Finally, patch merging and preprocessing yield the automatic HV and PV segmentation.

The strengths of D^2 -RD-UNet are manifold. The residual identity propagation aspect ensures efficient gradient flow, minimizing the risk of vanishing gradients during backpropagation and avoiding degradation problems [28]. This eases the training of deep networks, allowing the model to capture complex structures. Meanwhile, dense feature reuse enables each layer to use features from all preceding layers, enhancing feature richness and ensuring the network has access to multi-scale information [29]. This is particularly important for our task, where small structures such as vessels must be distinguished from surrounding tissues, even in low-resolution patches.

4.3.4. Output reconstruction

Both the 3D and 4D approaches generated multiclass predictions for HV, PV, and LV, perfectly aligned with the input patches. Since the inference operates on patches, the output remained fragmented. To address this challenge, we implemented a custom merging algorithm that handled overlapping regions by averaging voxel-wise predictions wherever overlaps occurred. Fig. 5 (bottom-right quadrant) depicts this repatching procedure. After reconstructing the full prediction over the liver-encompassing region, we used the spatial information retained during the localization phase to map the segmentation back into the coordinate space of the original CTA volume. This step ensured full spatial alignment with the input scan. We used the final segmentation mapped to the full CTA volume for all downstream evaluation metrics and postprocessing steps.

4.3.5. Loss function

While several advanced loss functions have been proposed to improve vascular segmentation by explicitly promoting topological

correctness, such as cDice [63], or topology-constrained loss, their effective integration required full-volume or wider field-of-view training. In a patch-based training setup, as adopted in this work due to GPU memory constraints, such losses would become less effective. In particular, cDice's soft skeletonization mechanism would fail when applied to topological interrupted patches, and topology-constrained losses could not enforce global vessel continuity without accessing to complete anatomical structures during optimization. Given these limitations, we combined Generalized Dice loss [64] and weighted Focal loss [65], balancing overlap-based and distribution-based learning signals. This choice enabled robust optimization even under patch constraints by simultaneously promoting regional accuracy and emphasizing misclassified examples. Our approach aligned with current best practices in recent vessel segmentation studies employing a combination of distribution-based loss and region-based loss [30,43,51–53, 66]. The mathematical formulation of our loss function can be found in Equation D.5 in Appendix D.1.

4.3.6. Global evaluation metrics

As mentioned in Section 1.2, a single metric like DSC is not enough for a comprehensive evaluation of automatic segmentation of vessels. To overcome this limitation, our evaluation method endowed metrics from different classes to provide a more comprehensive assessment. Additionally, we introduced the novel cDice metric proposed by [63] to quantify the centerlines discrepancies, ensuring a more precise evaluation of vessel continuity and topology preservation. We carefully selected a set of eight metrics presented in Table 4 to evaluate the network performances from different perspectives. Their complete mathematical formulation is reported in Appendix D.2.

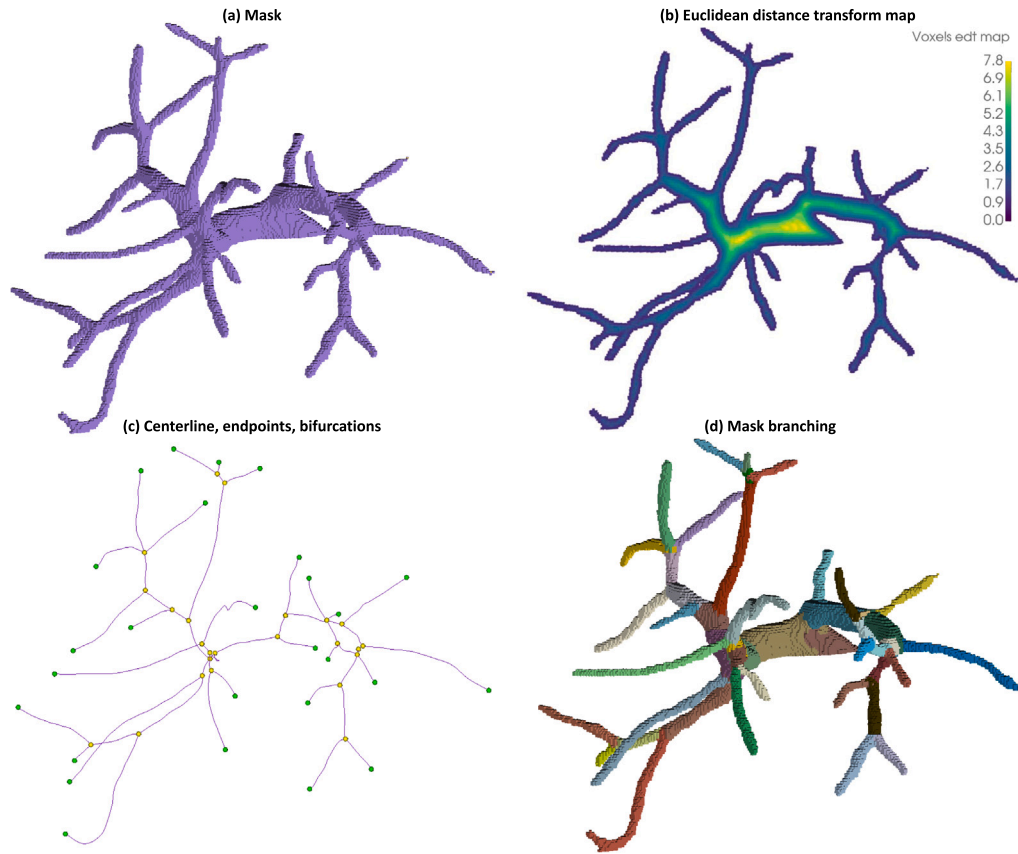


Fig. 6. Overview of our fully-automatic radius-based vessel branching approach. For visual clarity, the image was obtained by processing the GT mask instead of the common mask. (a) PV vascular tree of GT. (b) Euclidean Distance Transform (EDT) map of the PV structure. (c) The centerline shows endpoints (green) and bifurcations (yellow), with each segment linking them assigned to a mean radius computed from the EDT map. (d) PV tree is divided into branches, each marked by its specific mean radius. Each branch has a randomly assigned color.

Table 4
Global evaluation metrics.

Classes	Metrics	Definition
Region-based	Dice Score Coefficient (DSC)	Equation D.7
	Precision (PR)	Equation D.8
	Recall (RE)	Equation D.9
	Matthews Correlation Coefficient (MCC)	Equation D.10
Size-based	Relative Volume Difference (RVD)	Equation D.11
Distance-based	Hausdorff Distance (HD)	Equation D.12
	Average Symmetric Surface Distance (ASSD)	Equation D.14
Topology-based	Centerline DICE (cDice)	Equation D.18

4.3.7. Local evaluation metrics: radius-based approach

We developed a novel radius-based local evaluation pipeline to assess the quality of automatic reconstructions of HV and PV, focusing on the radii of the vascular branches. This method provides a detailed, branch-level comparison between predicted and GT vascular structures, addressing key limitations of existing global evaluation approaches. More specifically, the globally averaged metric does not equally weigh tubular structures with large, medium, and small radii [63]. The pipeline enables a localized evaluation of predicted vascular branches, categorized on their specific radius. This feature provides valuable insights into model behavior by highlighting accuracy across different radius ranges. This stratified evaluation is particularly helpful for preoperative planning, where the clinical relevance of vascular structures varies according to their size, location, and proximity to pathological regions such as tumors [67]. While smaller peripheral branches contribute to overall vascular completeness, their accuracy is

often less critical than larger primary and secondary branches, which are essential for surgical navigation and intervention. Additionally, vessels located near tumors are of paramount importance, as they directly influence surgical resection margins, the procedure's safety, and the risk of postoperative complications [68]. The radius-based evaluation aligns predictions with clinical priorities, enabling models to be assessed on their ability to achieve anatomical precision in key vascular regions, especially those near or involved in critical pathological sites.

The pipeline combines the GT and prediction masks into a common mask, creating a shared topology for evaluation. The process starts with an Euclidean Distance Transform (EDT) map applied to the common mask, retrieving the voxel-specific distance from the nearest background value, as shown in Fig. 6b. The skeleton extraction process generates a simplified representation of the vessel network by computing a line skeleton and refining it into a graph of vertices and edges. This skeleton is then associated with the EDT map by linking each vertex to its corresponding radius, derived from the EDT map values at specific spatial locations. This integration ensures that the skeleton captures the network's topology and reflects its spatial and structural properties, providing a robust foundation for quantitative analysis. Next, we identify endpoints and bifurcation points on the skeleton, as illustrated in Fig. 6c. We then integrate the skeleton's radius information with the classification of these points to segment the common mask into distinct branches, each with its average radius assigned using morphological operations (Fig. 6d). To ensure topological consistency between the GT and the prediction, we combine the common mask (logical OR) with both the GT and prediction masks, producing branch-segmented GT and prediction versions that are topologically overlapping and aligned for accurate comparison.

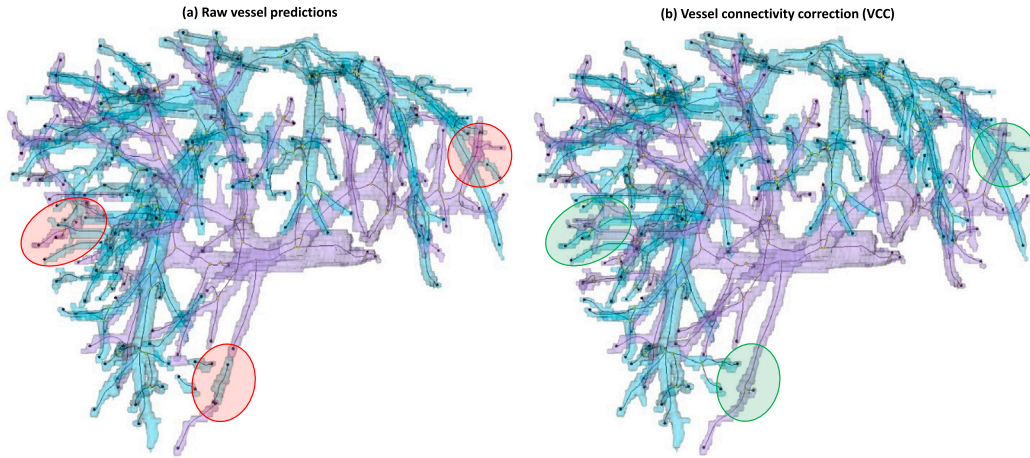


Fig. 7. Overview of our fully automated algorithm for vessel connectivity correction. (a) Raw vessel prediction shows misclassified regions, where some HV regions are incorrectly assigned to PV and vice versa. Examples of missing regions are highlighted in red. (b) Corrected PV and HV trees based on endpoint distance analysis. Examples of corrected regions are highlighted in green.

The branch-segmented GT and prediction are subsequently evaluated pairwise for each corresponding branch using the DSC, PR, RE, HD95, and RVD metrics, as detailed in Section 4.3.6. ASSD is omitted for its high computational cost on several fine-grained segmentations, while cDice is excluded due to unreliability on small branches.

This novel framework set a new benchmark for evaluating vascular reconstructions by enabling stratified accuracy assessment across radii. It provided actionable insights into model performance while aligning with the clinical demands of preoperative planning, where the precision of larger branches is a priority. By balancing methodological rigor with clinical relevance, this radius-specific evaluation ensured robust and clinically meaningful assessments of automatic reconstruction models.

4.4. Postprocessing

This section describes the preprocessing steps applied to the model output to enhance vessel structures and reduce noise. These steps include removing small outlier components and an algorithm designed to restore misclassified vessel connectivity.

4.4.1. Vessel connectivity correction

We designed the Vessel Connectivity Correction (VCC) algorithm to resolve frequent misclassification issues in multiclass vessel segmentation, which often interrupt the anatomical continuity of vascular trees by splitting them into disconnected components across different classes. After reconstructing the predicted labels within the full CTA volume space as described in Section 4.3.4, we applied the VCC algorithm to refine the segmentation results. It is important to note that VCC did not generate new vascular structures, but reassigned misclassified regions from the network. Therefore, its effectiveness relied on the network's ability to correctly identify bifurcation points and to distinguish well between vessels with very close crossings belonging to different classes. To clearly indicate that a model output has undergone postprocessing with VCC, we added the suffix -VCC to the model name.

The algorithm worked on HV and PV, which may consist of multiple connected components. A visual representation of the algorithm results is depicted in Fig. 7. As a preliminary refinement, the algorithm removed tiny connected components in both classes, identified as outliers, based on a statistical analysis of component size. This step reduced noise and improved the accuracy of subsequent processing by focusing on significant structures. After outliers removal, the centerlines were extracted for each connected component in both classes: Cnl_{HV} for HV and Cnl_{PV} for PV. Thus, the centerlines for the two classes were defined

as:

$$Cnl_{HV} = \{Cnl_1^{HV}, Cnl_2^{HV}, \dots, Cnl_m^{HV}\},$$

$$Cnl_{PV} = \{Cnl_1^{PV}, Cnl_2^{PV}, \dots, Cnl_n^{PV}\},$$

where Cnl_k^{HV} and Cnl_l^{PV} represent the individual centerlines for the k th connected component of HV and the l th connected component of PV, respectively. Since each centerline Cnl_k^{HV} and Cnl_l^{PV} can have more than two endpoints due to branching, the algorithm extracted all endpoints from each centerline. For each k th centerline of HV and l th centerline of PV, the endpoints were indexed and grouped as follows:

$$End_k^{HV} = \{e_{k1}^{HV}, e_{k2}^{HV}, \dots, e_{kp_k}^{HV}\},$$

$$End_l^{PV} = \{e_{l1}^{PV}, e_{l2}^{PV}, \dots, e_{lq_l}^{PV}\},$$

where e_{ki}^{HV} represents the i th endpoint of the k th centerline of HV, and e_{lj}^{PV} represents the j th endpoint of the l th centerline of PV. Here, p_k and q_l denote the total number of endpoints for the k th and l th centerlines in HV and PV, respectively. The algorithm then identified all pairs of endpoints (e_{ki}^{HV}, e_{lj}^{PV}) that belong to different class segments (i.e., one endpoint from HV and one from PV) and are proximal within a spherical kernel with radius $R = 4$, experimentally determined as optimal; the corresponding radii selection experiments on AIMS-HPV-385 TeS (considered as our clinical benchmark) are reported in Table 5. The mathematical formulation of the endpoints pair can be written as:

$$\|e_{ki}^{HV} - e_{lj}^{PV}\| \leq R = 4 \quad \text{for } e_{ki}^{HV} \in End_k^{HV}, \quad e_{lj}^{PV} \in End_l^{PV}$$

Once these endpoint pairs were identified, the algorithm reassigned the smaller associated segment while keeping the larger segment unchanged. This process was repeated iteratively until reaching a final configuration where no endpoint pairs of different classes remained, ensuring a more connected and accurate segmentation. For completeness, we noted that all observed cases of over-merging occurred in regions where the network failed to preserve class separation, typically producing continuous predictions between HV and PV branches. In these cases, VCC did not introduce new errors but propagates existing misclassifications to distal segments. The class-wise PR and RE values reported in Table 5 quantitatively supported this behavior, showing no systematic PR drop after applying VCC. This tendency indicated that the algorithm did not independently cause erroneous merging. Furthermore, the choice of $R = 4$ reflected the radius that maximized the cDice score, representing the best trade-off between restoring anatomical continuity and avoiding over-merging near bifurcations. These findings

Algorithm 1 Pseudocode of our methods for training and inference pipelines

```

1: 1. Liver Localization
2:   1.1 LiverRegion  $\leftarrow$  LocalizeLiver(CTA)  $\triangleright$  2.5D UNet
3:   1.2 Centroid  $\leftarrow$  GetCentroid(LiverRegion)
4: 2. Dual Pipeline
5:   2.1 If 3D pipeline:
6:     2.1.1 HUWindowed  $\leftarrow$  HUWindowing(LiverRegion,
7:       [55:275])
8:     2.1.2 Patches  $\leftarrow$  Patchify(HUWindowed, shape =
9:       (192,192,64,1))
10:    2.1.3 InputTensor  $\leftarrow$  Batch(Patches, B=4)
11:    2.1.4 LabelPatches  $\leftarrow$   $D^2$ -RD-UNet(InputTensor)
12:    2.1.5 LabelPatches.shape = (192,192,64,3)  $\triangleright$  HV, PV, LV
13:   2.2 If 4D pipeline:
14:     2.2.1 HUWindowed  $\leftarrow$  HUWindowing(LiverRegion,
15:       [55:275])
16:     2.2.2 Filters  $\leftarrow$  {Sato, Frangi, RORPO, OOF}
17:     2.2.3 FilteredVolumes  $\leftarrow$  ApplyFilters(LiverRegion, Fil-
18:       ters)
19:     2.2.4 HyperVolume  $\leftarrow$  Concatenate(HUWindowed, Fil-
20:       teredVolumes)
21:     2.2.5 Patches  $\leftarrow$  Patchify(HyperVolume, shape =
22:       (192,192,64,5))
23:     2.2.6 InputTensor  $\leftarrow$  Batch(Patches, B=4)
24:     2.2.7 LabelPatches  $\leftarrow$   $D^2$ -RD-UNet (4D)(InputTensor)
25:     2.2.8 LabelPatches.shape = (192,192,64,3)  $\triangleright$  HV, PV, LV
26: 3. Output Reconstruction
27:   3.1 LiverLabels  $\leftarrow$  MergePatches(LabelPatches, strategy =
28:     "average")
29:   3.2 FullLabels  $\leftarrow$  MapToFullVolume(LiverLabels, Centroid)
30: 4. Evaluation and Postprocessing
31:   4.1 If VCC enabled:
32:     4.1.1 FullLabels  $\leftarrow$  VCC(FullLabels)
33:     4.1.2 ModelName  $\leftarrow$  AppendSuffix("VCC")
34:   4.2 Evaluate(FullLabels)
35:   return FullLabels

```

confirmed the capability of VCC to enhance segmentation consistency without introducing systematic merging artifacts, particularly when used with the empirically validated radius of $R = 4$.

5. Experiments and discussion

We evaluated our models on the 58 subjects (TeS) from the AIMS-HPV-385 dataset. Then, we tested the generalization of our models on the 20 cases of the publicly available 3D-IRCADb-01 dataset. The evaluation procedure employed the comprehensive set of eight global metrics described in Section 4.3.6 on our 3D and 4D pipelines. We assessed our base D^2 -RD-UNet, also referred to as 3D, its variant D^2 -RD-UNet-VCC, which includes the vessel connectivity correction algorithm and also referred to as VCC, and finally D^2 -RD-UNet-VCC (4D), which integrates both the 4D setup and the VCC, for simplicity also referred to as 4D. To demonstrate the effectiveness of our approach, we benchmarked our models against four highly adopted SOTA architectures: nnU-Net, UNet-AG, and VNet, SwinUNETR.

5.1. Implementation

During the training phase on the AIMS-HPV-385 TrS, the models processed a total of 1860 patches, grouped with a batch size of 4 for both the 3D and 4D approaches. Each model underwent training for 50 epochs, using a learning rate of 0.0001. The training was implemented with the Tensorflow framework and 1 Nvidia A100 GPU with 40 GB

Table 5

Comparison of the median (IQR) values of DSC, Precision (PR), Recall (RE), and cDice for HV and PV structures across different VCC spherical kernel radii. The reported values are obtained using inference from the D^2 -RD-UNet without VCC (No VCC), as well as after applying VCC with different spherical kernel radii.

Hepatic Vein (HV)	DSC	PR	RE	cDice
No VCC	0.68 (0.10)	0.72 (0.10)	0.68 (0.17)	0.74 (0.09)
VCC ($R = 2$)	0.67 (0.10)	0.70 (0.10)	0.65 (0.15)	0.72 (0.08)
VCC ($R = 4$)	0.69 (0.11)	0.72 (0.11)	0.70 (0.17)	0.76 (0.09)
VCC ($R = 6$)	0.68 (0.09)	0.73 (0.09)	0.64 (0.15)	0.71 (0.08)
VCC ($R = 8$)	0.66 (0.10)	0.69 (0.10)	0.64 (0.15)	0.69 (0.08)
Portal Vein (PV)	DSC	PR	RE	cDice
No VCC	0.67 (0.08)	0.72 (0.10)	0.64 (0.15)	0.70 (0.10)
VCC ($R = 2$)	0.66 (0.10)	0.69 (0.10)	0.63 (0.15)	0.68 (0.08)
VCC ($R = 4$)	0.69 (0.08)	0.73 (0.10)	0.65 (0.15)	0.70 (0.11)
VCC ($R = 6$)	0.66 (0.09)	0.69 (0.10)	0.63 (0.15)	0.69 (0.08)
VCC ($R = 8$)	0.64 (0.10)	0.68 (0.10)	0.62 (0.15)	0.68 (0.08)

of memory, ensuring efficient computational performance. The total training time for the six different architectures, exploiting an efficient GPU data caching procedure amounts to 35.01 h.

5.2. Global metrics on AIMS-HPV-385 TeS

5.2.1. Quantitative results

The results reported in Table 6 show the greater adaptability of the D^2 -RD-UNet to the complexity of vascular segmentation than SOTA networks in three segmentation tasks.

For the HV segmentation task, the D^2 -RD-UNet framework outperformed nn-UNet, VNet and SwinUNETR in all eight metrics. Compared with UNet-AG, it scored higher in all the metrics except for precision, with a median of 0.72 vs. 0.76. Remarkably, we noted that the VCC algorithm did not provide an edge to the D^2 -RD-UNet for the HV segmentation task, as reflected by the lowest scores in all eight metrics. In fact, the 4D configuration achieved the highest median values among the others D^2 -RD-UNet variants in four metrics (DSC = 0.71, RE = 0.75, MCC = 0.71, and RVD = 0.01), the 3D version in one (ASSD = 1.37 mm), and the VCC version in two (cDice = 0.76, HD95 = 9.98 mm). In the remaining metric (PR), the three D^2 -RD-UNet variants reported the same score (PR = 0.72). For the PV segmentation task, our D^2 -RD-UNet framework outperformed the SOTA in all metrics. While the 3D version of D^2 -RD-UNet reported the highest values in two (HD95 = 13.26 mm, and ASSD = 1.76 mm) and the 4D version in four metrics (DSC = 0.70, RE = 0.71, cDice = 0.72, and RVD = -0.02), for this task the VCC algorithm was beneficial on 3D D^2 -RD-UNet-VCC, achieving the highest PR (0.73) and MCC (0.70) (Table 6). For the UV segmentation task, the D^2 -RD-UNet framework performed better than SOTA in all metrics (Table 6), with the 4D on four (DSC = 0.72, RE = 0.75, cDice = 0.78, and RVD = 0.0), the 3D on one (PR = 0.74). The VCC achieved the highest values on four metrics (PR = 0.74 as the 3D version, MCC = 0.72, HD95 = 7.10 mm, and ASD = 1.11 mm).

The increments achieved by the D^2 -RD-UNet framework underscored the network's ability to accurately identify vessel centerlines, which is critical for both clinical applications and maintaining the integrity of the overall vascular topology. Overall, the 4D configuration, which combined CT raw input with vesselness filters (Sato, Frangi, RORPO, and OOF), was the configuration that improved most of the values of the segmentation metrics across all vascular structures. This enhancement is particularly evident for HV, where D^2 -RD-UNet (4D) reduced HD95 by 0.32 mm compared to nnU-Net (10.30 mm vs. 9.98 mm) and ASSD by 0.30 mm (1.67 mm vs. 1.37 mm). For PV, D^2 -RD-UNet (4D) reduced ASSD by 0.18 mm compared to nnU-Net (2.17 mm vs. 1.99 mm) and by 0.67 mm compared to VNet (2.66 mm vs. 1.99 mm). Interestingly, the transformer-based SwinUNETR model does not deviate significantly from the performance of CNN-based models (nnU-Net, VNet, UNet-AG), showing slight improvements in

Table 6

AIMS-385-HPV TeS quantitative evaluation of vessel segmentation performances. The table presents a comparative analysis of segmentation performance for HV, PV, and UV across SOTA methods (nnU-Net, VNet, U-Net-AG, SwinUNETR) and our three D^2 -RD-UNet variants. The evaluation is conducted on the AIMS-385-HPV TeS, consisting of **58 held-out subjects**. Results are reported as Mean $\pm$ Std and Median (IQR) across eight metrics. The best values are highlighted in bold.

Hepatic Vessel (HV)	Metric	DSC	PR	RE	MCC	clDice	HD95 [mm]	ASSD [mm]	RVD
nnU-Net [69]	Mean $\pm$ Std	0.62 $\pm$ 0.14	0.64 $\pm$ 0.12	0.63 $\pm$ 0.17	0.62 $\pm$ 0.13	0.67 $\pm$ 0.17	15.68 $\pm$ 17.29	2.97 $\pm$ 4.70	-0.01 $\pm$ 0.29
	Median (IQR)	0.66 (0.09)	0.67 (0.12)	0.66 (0.13)	0.66 (0.08)	0.72 (0.12)	10.30 (4.32)	1.67 (0.71)	0.03 (0.22)
	(Min, Max)	(0.36, 0.76)	(0.34, 0.80)	(0.25, 0.85)	(0.36, 0.76)	(0.32, 0.84)	(7.00, 31.88)	(0.98, 5.10)	(-0.60, 0.53)
VNet [70]	Mean $\pm$ Std	0.60 $\pm$ 0.16	0.70 $\pm$ 0.11	0.55 $\pm$ 0.17	0.60 $\pm$ 0.15	0.64 $\pm$ 0.17	18.72 $\pm$ 23.02	4.78 $\pm$ 12.21	-0.19 $\pm$ 0.26
	Median (IQR)	0.64 (0.12)	0.71 (0.11)	0.59 (0.16)	0.64 (0.11)	0.68 (0.10)	12.39 (6.60)	1.88 (1.24)	-0.15 (0.28)
	(Min, Max)	(0.25, 0.75)	(0.45, 0.85)	(0.15, 0.77)	(0.32, 0.75)	(0.28, 0.83)	(6.40, 35.75)	(0.94, 11.16)	(-0.78, 0.21)
UNet-AG [71]	Mean $\pm$ Std	0.60 $\pm$ 0.15	0.74 $\pm$ 0.12	0.53 $\pm$ 0.16	0.61 $\pm$ 0.14	0.61 $\pm$ 0.16	21.42 $\pm$ 22.15	5.06 $\pm$ 11.59	-0.26 $\pm$ 0.24
	Median (IQR)	0.65 (0.12)	0.76 (0.12)	0.57 (0.14)	0.66 (0.12)	0.64 (0.12)	15.02 (5.80)	2.47 (1.09)	-0.24 (0.23)
	(Min, Max)	(0.26, 0.76)	(0.45, 0.92)	(0.17, 0.78)	(0.32, 0.76)	(0.34, 0.79)	(9.00, 50.10)	(1.31, 9.29)	(-0.74, 0.32)
SwinUNETR [72]	Mean $\pm$ Std	0.58 $\pm$ 0.14	0.67 $\pm$ 0.13	0.54 $\pm$ 0.16	0.60 $\pm$ 0.13	0.62 $\pm$ 0.15	20.04 $\pm$ 12.91	3.64 $\pm$ 3.25	-0.16 $\pm$ 0.33
	Median (IQR)	0.62 (0.16)	0.70 (0.14)	0.58 (0.20)	0.64 (0.14)	0.65 (0.15)	16.40 (13.78)	2.55 (2.05)	-0.17 (0.36)
	(Min, Max)	(0.13, 0.73)	(0.14, 0.87)	(0.08, 0.82)	(0.22, 0.73)	(0.09, 0.84)	(5.48, 82.88)	(1.10, 18.00)	(-0.88, 1.32)
D^2 -RD-UNet (Ours)	Mean $\pm$ Std	0.65 $\pm$ 0.15	0.70 $\pm$ 0.12	0.63 $\pm$ 0.17	0.65 $\pm$ 0.13	0.72 $\pm$ 0.13	15.53 $\pm$ 26.96	4.15 $\pm$ 14.96	-0.11 $\pm$ 0.28
	Median (IQR)	0.68 (0.10)	0.72 (0.10)	0.68 (0.17)	0.69 (0.09)	0.74 (0.09)	10.02 (4.71)	1.37 (0.60)	-0.05 (0.27)
	(Min, Max)	(0.36, 0.79)	(0.54, 0.86)	(0.24, 0.83)	(0.41, 0.79)	(0.39, 0.87)	(4.12, 28.65)	(0.88, 4.89)	(-0.66, 0.41)
D^2 -RD-UNet-VCC (Ours)	Mean $\pm$ Std	0.66 $\pm$ 0.15	0.69 $\pm$ 0.13	0.64 $\pm$ 0.18	0.68 $\pm$ 0.14	0.72 $\pm$ 0.14	16.81 $\pm$ 27.49	4.49 $\pm$ 15.05	-0.10 $\pm$ 0.27
	Median (IQR)	0.69 (0.11)	0.72 (0.11)	0.70 (0.17)	0.70 (0.11)	0.76 (0.09)	9.98 (6.77)	1.44 (0.81)	-0.05 (0.28)
	(Min, Max)	(0.44, 0.79)	(0.54, 0.85)	(0.31, 0.85)	(0.49, 0.79)	(0.38, 0.87)	(3.32, 43.19)	(0.82, 9.66)	(-0.59, 0.41)
D^2 -RD-UNet-VCC (4D) (Ours)	Mean $\pm$ Std	0.68 $\pm$ 0.12	0.71 $\pm$ 0.1	0.68 $\pm$ 0.17	0.68 $\pm$ 0.11	0.72 $\pm$ 0.13	20.43 $\pm$ 25.38	3.35 $\pm$ 5.77	-0.01 $\pm$ 0.29
	Median (IQR)	0.71 (0.12)	0.72 (0.10)	0.75 (0.17)	0.71 (0.12)	0.76 (0.10)	11.02 (12.38)	1.63 (1.20)	0.01 (0.20)
	(Min, Max)	(0.28, 0.83)	(0.5, 0.94)	(0.16, 0.9)	(0.36, 0.83)	(0.22, 0.84)	(3.16, 130.28)	(0.81, 37.45)	(-0.82, 0.79)
Portal Vessel (PV)	Metric	DSC	PR	RE	MCC	clDice	HD95 [mm]	ASSD [mm]	RVD
nnU-Net [69]	Mean $\pm$ Std	0.62 $\pm$ 0.12	0.67 $\pm$ 0.10	0.61 $\pm$ 0.17	0.63 $\pm$ 0.10	0.63 $\pm$ 0.15	20.27 $\pm$ 19.66	3.61 $\pm$ 4.87	-0.06 $\pm$ 0.33
	Median (IQR)	0.64 (0.09)	0.67 (0.12)	0.62 (0.14)	0.65 (0.08)	0.66 (0.10)	14.95 (9.08)	2.17 (1.25)	-0.07 (0.33)
	(Min, Max)	(0.24, 0.81)	(0.46, 0.86)	(0.14, 0.90)	(0.32, 0.81)	(0.19, 0.88)	(2.00, 135.04)	(0.84, 13.64)	(-0.80, 0.80)
VNet [70]	Mean $\pm$ Std	0.60 $\pm$ 0.13	0.68 $\pm$ 0.10	0.58 $\pm$ 0.16	0.61 $\pm$ 0.11	0.60 $\pm$ 0.14	22.46 $\pm$ 22.33	4.46 $\pm$ 8.09	-0.11 $\pm$ 0.32
	Median (IQR)	0.62 (0.06)	0.68 (0.12)	0.59 (0.13)	0.63 (0.07)	0.61 (0.10)	16.17 (8.38)	2.66 (1.24)	-0.13 (0.28)
	(Min, Max)	(0.27, 0.79)	(0.45, 0.85)	(0.17, 0.93)	(0.34, 0.79)	(0.25, 0.87)	(5.74, 134.90)	(0.93, 12.09)	(-0.75, 0.93)
UNet-AG [71]	Mean $\pm$ Std	0.61 $\pm$ 0.13	0.67 $\pm$ 0.11	0.59 $\pm$ 0.15	0.61 $\pm$ 0.11	0.60 $\pm$ 0.14	19.24 $\pm$ 15.73	4.17 $\pm$ 8.10	-0.09 $\pm$ 0.30
	Median (IQR)	0.62 (0.08)	0.68 (0.12)	0.60 (0.15)	0.63 (0.08)	0.62 (0.11)	14.28 (5.50)	2.42 (0.75)	-0.12 (0.25)
	(Min, Max)	(0.32, 0.81)	(0.36, 0.86)	(0.22, 0.85)	(0.32, 0.81)	(0.29, 0.86)	(5.71, 52.39)	(0.77, 8.60)	(-0.70, 0.75)
SwinUNETR [72]	Mean $\pm$ Std	0.60 $\pm$ 0.10	0.67 $\pm$ 0.11	0.57 $\pm$ 0.14	0.61 $\pm$ 0.09	0.59 $\pm$ 0.13	24.05 $\pm$ 13.24	4.07 $\pm$ 3.01	-0.12 $\pm$ 0.28
	Median (IQR)	0.61 (0.10)	0.67 (0.12)	0.57 (0.15)	0.62 (0.10)	0.60 (0.14)	19.47 (13.84)	3.02 (2.59)	-0.15 (0.24)
	(Min, Max)	(0.23, 0.82)	(0.34, 0.91)	(0.14, 0.93)	(0.32, 0.82)	(0.15, 0.87)	(3.16, 73.83)	(0.75, 17.33)	(-0.81, 0.71)
D^2 -RD-UNet (Ours)	Mean $\pm$ Std	0.65 $\pm$ 0.13	0.71 $\pm$ 0.10	0.63 $\pm$ 0.16	0.66 $\pm$ 0.11	0.67 $\pm$ 0.15	17.76 $\pm$ 16.30	3.59 $\pm$ 7.24	-0.10 $\pm$ 0.28
	Median (IQR)	0.67 (0.08)	0.72 (0.10)	0.64 (0.15)	0.68 (0.08)	0.70 (0.10)	13.26 (8.11)	1.76 (1.18)	-0.12 (0.26)
	(Min, Max)	(0.25, 0.81)	(0.43, 0.89)	(0.16, 0.89)	(0.33, 0.81)	(0.27, 0.90)	(4.36, 55.79)	(0.69, 16.17)	(-0.77, 0.54)
D^2 -RD-UNet-VCC (Ours)	Mean $\pm$ Std	0.67 $\pm$ 0.13	0.72 $\pm$ 0.10	0.63 $\pm$ 0.16	0.68 $\pm$ 0.11	0.68 $\pm$ 0.15	19.23 $\pm$ 16.90	3.70 $\pm$ 7.23	-0.10 $\pm$ 0.29
	Median (IQR)	0.69 (0.08)	0.73 (0.10)	0.65 (0.15)	0.70 (0.08)	0.70 (0.11)	13.41 (12.25)	1.78 (1.68)	-0.11 (0.26)
	(Min, Max)	(0.26, 0.80)	(0.43, 0.90)	(0.16, 0.90)	(0.33, 0.80)	(0.28, 0.88)	(4.12, 52.66)	(0.69, 15.48)	(-0.77, 0.66)
D^2 -RD-UNet-VCC (4D) (Ours)	Mean $\pm$ Std	0.68 $\pm$ 0.1	0.70 $\pm$ 0.1	0.69 $\pm$ 0.14	0.69 $\pm$ 0.08	0.69 $\pm$ 0.13	22.53 $\pm$ 20.82	3.59 $\pm$ 5.91	0.02 $\pm$ 0.29
	Median (IQR)	0.70 (0.09)	0.71 (0.12)	0.71 (0.13)	0.70 (0.09)	0.72 (0.11)	15.59 (10.94)	1.99 (1.68)	-0.02 (0.24)
	(Min, Max)	(0.25, 0.84)	(0.48, 0.94)	(0.14, 0.95)	(0.36, 0.84)	(0.12, 0.86)	(2.24, 112.48)	(0.42, 42.74)	(-0.85, 0.7)
Unified Vessel (UV)	Metric	DSC	PR	RE	MCC	clDice	HD95 [mm]	ASSD [mm]	RVD
nnU-Net [69]	Mean $\pm$ Std	0.66 $\pm$ 0.10	0.69 $\pm$ 0.08	0.65 $\pm$ 0.14	0.66 $\pm$ 0.08	0.72 $\pm$ 0.13	10.93 $\pm$ 11.79	2.14 $\pm$ 3.71	-0.03 $\pm$ 0.25
	Median (IQR)	0.68 (0.06)	0.69 (0.11)	0.68 (0.14)	0.68 (0.05)	0.75 (0.08)	7.55 (3.96)	1.22 (0.71)	-0.01 (0.24)
	(Min, Max)	(0.49, 0.78)	(0.54, 0.85)	(0.37, 0.86)	(0.50, 0.78)	(0.52, 0.86)	(4.12, 30.30)	(0.89, 4.24)	(-0.56, 0.45)
VNet [70]	Mean $\pm$ Std	0.64 $\pm$ 0.13	0.74 $\pm$ 0.08	0.60 $\pm$ 0.15	0.65 $\pm$ 0.11	0.69 $\pm$ 0.15	13.37 $\pm$ 16.30	3.13 $\pm$ 8.12	-0.17 $\pm$ 0.23
	Median (IQR)	0.66 (0.07)	0.73 (0.11)	0.62 (0.14)	0.67 (0.06)	0.73 (0.08)	8.91 (4.73)	1.44 (0.87)	-0.15 (0.24)
	(Min, Max)	(0.44, 0.78)	(0.57, 0.86)	(0.33, 0.75)	(0.48, 0.78)	(0.48, 0.84)	(3.32, 37.10)	(0.78, 6.08)	(-0.57, 0.21)
UNet-AG [71]	Mean $\pm$ Std	0.64 $\pm$ 0.13	0.75 $\pm$ 0.08	0.59 $\pm$ 0.14	0.65 $\pm$ 0.11	0.68 $\pm$ 0.15	13.61 $\pm$ 14.92	3.12 $\pm$ 7.60	-0.21 $\pm$ 0.22
	Median (IQR)	0.68 (0.06)	0.75 (0.12)	0.62 (0.13)	0.68 (0.07)	0.71 (0.09)	9.25 (4.63)	1.46 (0.78)	-0.18 (0.24)
	(Min, Max)	(0.43, 0.77)	(0.57, 0.90)	(0.31, 0.80)	(0.47, 0.77)	(0.43, 0.85)	(4.12, 30.10)	(0.83, 5.27)	(-0.58, 0.27)
SwinUNETR [72]	Mean $\pm$ Std	0.64 $\pm$ 0.10	0.72 $\pm$ 0.09	0.59 $\pm$ 0.13	0.64 $\pm$ 0.08	0.68 $\pm$ 0.13	11.98 $\pm$ 6.45	2.15 $\pm$ 1.88	-0.17 $\pm$ 0.22
	Median (IQR)	0.67 (0.09)	0.73 (0.13)	0.62 (0.12)	0.67 (0.08)	0.71 (0.10)	10.05 (5.63)	1.51 (1.00)	-0.15 (0.27)
	(Min, Max)	(0.26, 0.76)	(0.51, 0.90)	(0.15, 0.78)	(0.37, 0.76)	(0.19, 0.85)	(5.10, 40.62)	(0.94, 12.74)	(-0.83, 0.31)
D^2 -RD-UNet (Ours)	Mean $\pm$ Std	0.67 $\pm$ 0.11	0.74 $\pm$ 0.08	0.64 $\pm$ 0.14	0.68 $\pm$ 0.10	0.73 $\pm$ 0.14	11.59 $\pm$ 15.38	2.58 $\pm$ 7.04	-0.11 $\pm$ 0.23
	Median (IQR)	0.69 (0.07)	0.74 (0.09)	0.68 (0.15)	0.70 (0.06)	0.76 (0.08)	7.62 (4.30)	1.15 (0.55)	-0.08 (0.26)
	(Min, Max)	(0.45, 0.78)	(0.56, 0.87)	(0.34, 0.82)	(0.48, 0.78)	(0.47, 0.87)	(3.46, 28.25)	(0.83, 6.16)	(-0.55, 0.30)
D^2 -RD-UNet-VCC (Ours)	Mean $\pm$ Std	0.68 $\pm$ 0.11	0.74 $\pm$ 0.08	0.65 $\pm$ 0.14	0.69 $\pm$ 0.10	0.74 $\pm$ 0.13	11.23 $\pm$ 15.46	2.52 $\pm$ 7.04	-0.10 $\pm$ 0.23
	Median (IQR)	0.71 (0.07)	0.74 (0.09)	0.69 (0.15)	0.72 (0.06)	0.77 (0.06)	7.10 (4.15)	1.11 (0.54)	-0.08 (0.25)
	(Min, Max)	(0.45, 0.78)	(0.56, 0.87)	(0.34, 0.83)	(0.48, 0.79)	(0.47, 0.87)	(3.61, 27.90)	(0.76, 6.07)	(-0.50, 0.31)
D^2 -RD-UNet-VCC (4D) (Ours)	Mean $\pm$ Std	0.70 $\pm$ 0.1	0.73 $\pm$ 0.09	0.71 $\pm$ 0.14	0.71 $\pm$ 0.08	0.75 $\pm$ 0.12	12.73 $\pm$ 18.69	2.09 $\pm$ 3.78	-0.01 $\pm$ 0.25
	Median (IQR)	0.72 (0.09)	0.72 (0.10)	0.75 (0.10)	0.72 (0.08)	0.78 (0.08)	7.81 (4.35)	1.20 (0.51)	0.0 (0.22)
	(Min, Max)	(0.27, 0.82)	(0.53, 0.96)	(0.16, 0.88)	(0.39, 0.82)	(0.18, 0.85)	(3.16, 108.22)	(0.71, 27.4)	(-0.84, 0.66)

some metrics. For **HV**, SwinUNETR achieves a higher RE (0.58 vs 0.57), a higher cDice (0.65 vs 0.64), and a lower RVD (−0.17 vs −0.24) compared to UNet-AG. For **PV**, SwinUNETR matches nnU-Net in PR, both scoring 0.67. In the **UV**, SwinUNETR outperforms VNet with a higher DSC (0.67 vs 0.66), a higher PR (0.73 vs 0.69), and outperforms UNet-AG with a lower RVD (−0.15 vs −0.18). Additionally, for **UV**, SwinUNETR matches VNet in PR (0.73), RE (0.62), and MCC (0.67), and matches UNet-AG in RE (0.62) and cDice (0.71). The improvements in boundary metrics reported by D^2 -RD-UNet framework highlighted its power to produce precise boundary delineations. Our analysis also revealed challenges inherent to vascular segmentation. PR tended to exceed RE for most architectures, including our baseline D^2 -RD-UNet, reflecting a tendency to produce false positives caused by the misclassification of vascular segments between HV and PV. This aspect highlighted the challenge of distinguishing between the two structures with similar HU distributions while preserving continuous topology. This issue was predominantly observed in closely intertwined vascular structures. In contrast with our initial hypothesis regarding patch boundary artifacts, errors were also evident in the central regions of the patches, suggesting that the primary contributing factor was the complexity of the vascular structures rather than boundary artifacts. Our D^2 -RD-UNet-VCC integrated a VCC algorithm to address this, significantly improving RE while maintaining PR. For example, VCC increased RE for HV by 6% compared to D^2 -RD-UNet (0.70 vs 0.64), with PR remaining consistent at 0.72. In UV segmentation, VCC reduced HD95 by 0.52 mm compared to D^2 -RD-UNet (7.62 mm vs. 7.10 mm) and ASSD by 0.04 mm (1.15 mm vs. 1.11 mm). These improvements have demonstrated the effectiveness of VCC in mitigating false positives and enhancing overall segmentation quality. Interestingly, the D^2 -RD-UNet (4D) configuration exhibited a distinct trade-off between PR and RE, favoring higher recall. For instance, RE reaches 0.75 in HV compared to PR at 0.72, suggesting that the network prioritizes sensitivity to finer vascular structures. While this trade-off enhanced the detection of thin and branching vessels, it also highlighted the need for further optimization to achieve a better balance between precision and recall.

In summary, the increments achieved by the D^2 -RD-UNet framework underscored the network's ability to accurately identify vessel centerlines, which is critical for both clinical applications and maintaining the integrity of the overall vascular topology. Additionally, the D^2 -RD-UNet (4D) and D^2 -RD-UNet-VCC achieved consistent and significant improvements across all vascular structures and evaluation metrics. These results highlighted the robustness and adaptability of our approach to the unique challenges of vascular segmentation.

5.2.2. Qualitative results

The qualitative evaluation of the reconstructed 3D vascular structures from the 58 test subjects in the AIMS-HPV-385 dataset revealed several notable strengths of the D^2 -RD-UNet variants, particularly those enhanced by the VCC and vesselness filter 4D approaches compared to the GT. A visual representation of two sample subjects is illustrated in Fig. 8. These models show superior continuity and detail in capturing the finer branches of both HV and PV, ensuring fine topology preservation and anatomically accurate representation of the vascular trees. The spatial coherence achieved by our models ensures that even in regions of complex bifurcations, branches remain connected and consistent, reducing gaps or discontinuities prevalent in other architectures. The qualitative improvements of the D^2 -RD-UNet variants are further evident in the pathological subject shown in Fig. 8(2). The models performed well in pathological scenarios, demonstrating the robustness of the training dataset in capturing and generalizing complex vascular geometries even under challenging and pathological conditions. A notable advantage of the D^2 -RD-UNet models, especially with VCC, was their ability to minimize false positives and reduce noise in areas surrounding the vessels, enhancing the overall quality and interpretability of the reconstructions. Integrating vesselness-filters 4D inputs further enriched these models' ability to accurately distinguish

small vessel branches and intricate details often lost in models like VNet and nnU-Net. Although SwinUNETR did not qualitatively deviate significantly from the performance of CNN-based models, it maintained good generalizability between physiological and pathological cases in the AIMS-HPV-385 dataset. However, our D^2 -RD-UNet variants continued to perform better in terms of topological accuracy and continuity. Overall, these qualitative observations emphasized the robustness of D^2 -RD-UNet in providing high-fidelity vascular reconstructions. These results made the method well-suited for 3D visualization for surgical planning, where anatomical precision and reliability are essential [8–11].

5.3. Global metrics on 3D-IRCADb-01

5.3.1. Quantitative results

The results on the 3D-IRCADb-01 dataset, including 20 healthy and pathological subjects, are reported in Table 7. Though widely used for benchmarking, this dataset presents specific challenges due to the less detailed public GT, as mentioned in Section 3.2. Evaluating the models on such GT masks introduced the expectation of overestimated errors. This observation was primarily related to the potential for over-segmentation compared to the more detailed and capillary-level annotations of AIMS-HPV-385 TrS. Since the networks are trained on the more detailed AIMS-HPV-385 TrS labels, the resulting predictions may be more precise than the GT masks of the 3D-IRCADb-01 test set. This mismatch in label quality can result in worse evaluation metrics on the 3D-IRCADb-01 test set, as the improved predictions may extend beyond the less accurate 3D-IRCADb-01 GT boundaries, creating an apparent over-segmentation effect. Despite these limitations, the D^2 -RD-UNet framework consistently demonstrated superior performance to SOTA networks in automatically segmenting HV, PV, and UV.

For the **HV** segmentation task, the D^2 -RD-UNet framework outperformed VNet and SwinUNETR in all eight metrics. Compared with UNet-AG, 3D and VCC configuration scored higher in all the metrics except for precision, with a median of 0.72 vs. 0.74, and the same for 4D configuration, with a median of 0.70 vs 0.74. Compared to nnU-Net, the 3D, 4D, and VCC configurations achieved higher scores across all metrics, except for RVD, whose median values were −0.24, −0.31, and −0.21, respectively, vs 0.14. The VCC configuration achieved the highest median values among the 3D and 4D variants in three metrics (DSC = 0.62, RE = 0.56, ASSD = 2.90), while the 3D version in one (HD95 = 15.00 mm), and 4D version in three (DSC = 0.62 same as VCC, MCC = 0.63, cDice = 0.74). For the **PV** segmentation task, SwinUNETR outperformed all three D^2 -RD-UNet variants in PR, achieving the highest median value of 0.70 among all models. Our D^2 -RD-UNet framework outperformed nnU-Net and UNet-AG in all eight metrics. The D^2 -RD-UNet 3D configuration reported the highest value in one metric (RVD = 0.00). The D^2 -RD-UNet with VCC algorithm was notably valuable for this task, achieving the highest values for four metrics (DSC = 0.68, PR = 0.69, MCC = 0.69, HD95 = 16.72 mm). The D^2 -RD-UNet in 4D configuration reported the highest values in two metrics (RE = 0.72, ASSD = 2.57 mm). For the **UV** segmentation task, the D^2 -RD-UNet framework performed better than SOTA in six metrics (DSC, RE, MCC, cDice, HD95, ASSD). 3D and VCC configurations scored higher in DSC, RE, cDice, HD95, and ASSD than the SOTA network, while lower in PR, MCC, and RVD. Likewise, the 4D version outperformed the SOTA network across all metrics except PR and RVD. Similar to AIMS-HPV-385 TeS, SwinUNETR achieved similar performance compared to CNN-based models across all vessel types. For the **HV**, it outperformed VNet in DSC (0.56 vs 0.55), RE (0.51 vs 0.48), and also UNet-AG in RE (0.51 vs 0.49) and cDice (0.65 vs 0.64). In **PV**, SwinUNETR scored higher than all three CNN baselines in DSC (0.62 vs 0.62 in nnU-Net, 0.60 in VNet and UNet-AG), PR (0.69 vs 0.62 in nnU-Net and UNet-AG), and cDice (0.63 vs 0.62 in VNet, 0.60 in UNet-AG). Additionally, SwinUNETR matched nnU-Net in RVD (−0.11). For **UV**, it reported the same performance as VNet in DSC (0.62) and MCC (0.63),

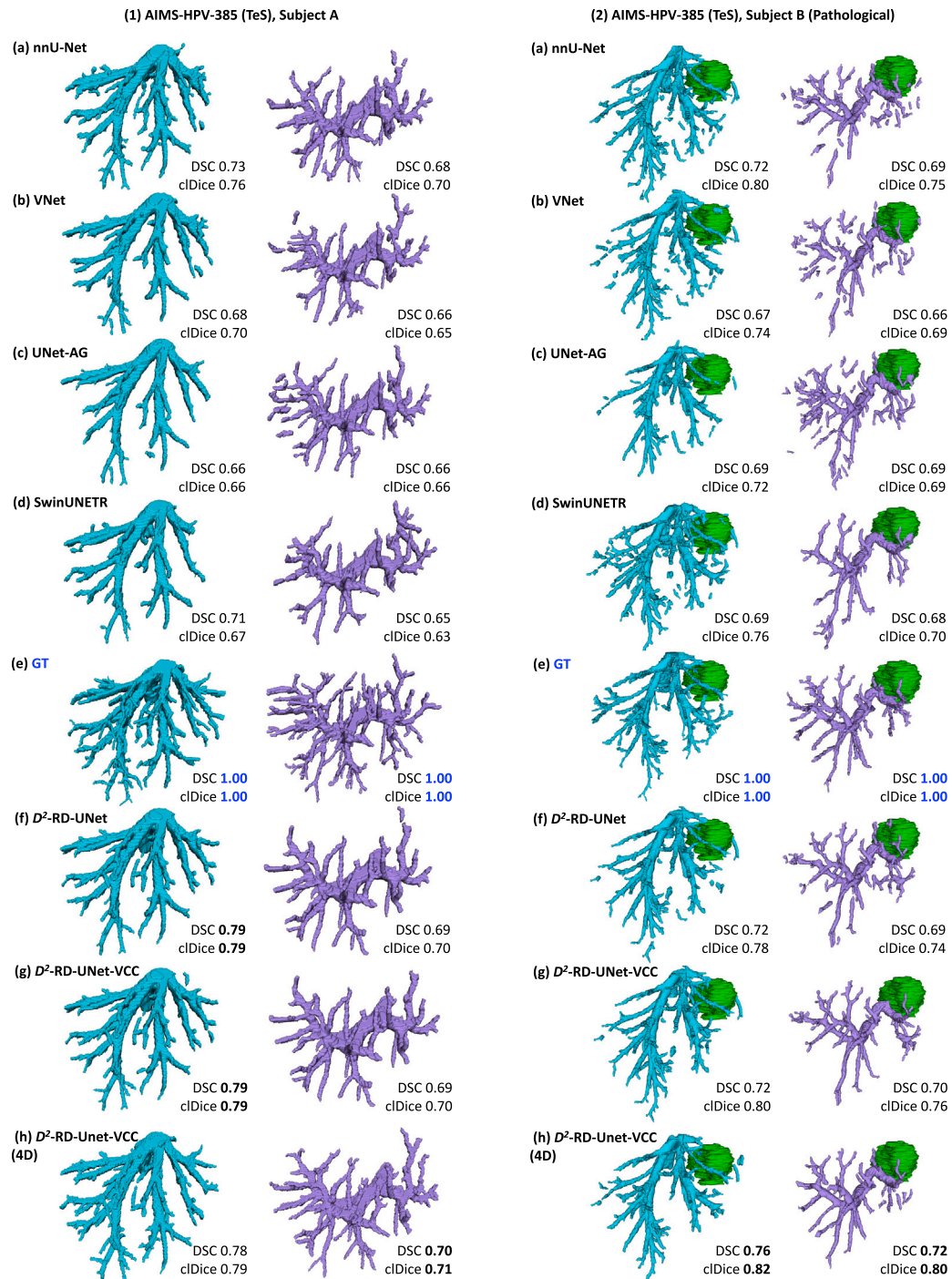


Fig. 8. AIMS-385-HPV TeS qualitative comparison of vessel segmentation methods. The figure shows HV (cyan) and PV (purple) segmentations for two subjects: (1) Subject A and (2) Subject B, a pathological case with the tumor manually segmented by an expert radiologist. Methods compared include SOTA models (nnU-Net, VNet, UNet-AG, SwinUNETR) and our three D^2 -RD-UNet variants. (e) shows the GT segmentations for reference. DSC and cDice are reported for each method, with the best values highlighted in bold.

while outperforming VNet in RE (0.58 vs 0.56), and UNet-AG in ASSD (2.33 mm vs 2.24 mm) and RVD (−0.22 vs −0.24).

Interestingly, our results showed that the proposed D^2 -RD-UNet optimized CNN-based framework consistently outperformed SwinUNETR across both test datasets. Statistical analysis using the Wilcoxon rank sum test confirmed that these differences were significant in most cases. On the AIMS-HPV-385 TeS, the performance gap was especially evident: for hepatic veins (HV), the difference was statistically significant with $p = 0.0001$, and for portal veins (PV) with $p = 0.00001$. On the smaller 3D-IRCADb-01 dataset, although the test set size made it more

challenging to reach statistical significance, our model still showed higher median performance than SwinUNETR. Specifically, for HV, the difference was not statistically significant ($p = 0.08$), though the median Dice scores are 0.60 (Ours) vs. 0.56 (SwinUNETR); for PV, the improvement was statistically significant ($p = 0.01$), with median Dice scores of 0.68 (ours) vs. 0.62 (SwinUNETR). These results corroborated the observation by Isensee et al. [73] that “most methods introduced in recent years fail to surpass the original nnU-Net baseline introduced in 2018”. Their analysis suggested that, despite growing interest in

Table 7

3D-IRCADb-01 quantitative evaluation of vessel segmentation performances. The table presents a comparative analysis of segmentation performance for HV, PV, and UV across SOTA methods (nnU-Net, VNet, U-Net-AG, SwinUNETR) and our three D^2 -RD-UNet variants. The evaluation is conducted on the whole 3D-IRCADb-01, consisting of **20 held-out subjects**. Results are reported as Mean $\pm$ Std and Median (IQR) across eight metrics. The best values are highlighted in bold.

Hepatic Vessel (HV)	Metric	DSC	PR	RE	MCC	clDice	HD95 [mm]	ASSD [mm]	RVD
nnU-Net [69]	Mean $\pm$ Std	0.56 $\pm$ 0.11	0.63 $\pm$ 0.15	0.55 $\pm$ 0.15	0.58 $\pm$ 0.09	0.66 $\pm$ 0.12	24.19 $\pm$ 17.73	4.03 $\pm$ 2.21	-0.05 $\pm$ 0.44
	Median (IQR)	0.58 (0.15)	0.64 (0.15)	0.57 (0.17)	0.58 (0.10)	0.68 (0.10)	19.26 (11.64)	3.44 (1.96)	-0.14 (0.39)
	(Min, Max)	(0.36, 0.73)	(0.33, 0.90)	(0.24, 0.86)	(0.42, 0.74)	(0.33, 0.82)	(9.00, 72.94)	(1.81, 10.93)	(-0.67, 0.96)
VNet [70]	Mean $\pm$ Std	0.53 $\pm$ 0.11	0.70 $\pm$ 0.16	0.47 $\pm$ 0.15	0.56 $\pm$ 0.09	0.63 $\pm$ 0.14	26.69 $\pm$ 22.08	4.80 $\pm$ 3.68	-0.27 $\pm$ 0.35
	Median (IQR)	0.55 (0.14)	0.68 (0.20)	0.48 (0.17)	0.55 (0.13)	0.66 (0.15)	18.94 (12.19)	3.76 (2.33)	-0.29 (0.34)
	(Min, Max)	(0.31, 0.69)	(0.38, 0.95)	(0.19, 0.77)	(0.41, 0.69)	(0.20, 0.81)	(9.27, 87.78)	(1.62, 17.51)	(-0.80, 0.50)
UNet-AG [71]	Mean $\pm$ Std	0.54 $\pm$ 0.11	0.73 $\pm$ 0.15	0.47 $\pm$ 0.15	0.57 $\pm$ 0.09	0.62 $\pm$ 0.13	24.44 $\pm$ 16.18	4.39 $\pm$ 2.24	-0.29 $\pm$ 0.37
	Median (IQR)	0.57 (0.06)	0.74 (0.16)	0.49 (0.15)	0.58 (0.08)	0.64 (0.12)	19.84 (10.15)	3.81 (2.29)	-0.34 (0.43)
	(Min, Max)	(0.27, 0.70)	(0.40, 0.97)	(0.17, 0.77)	(0.36, 0.70)	(0.30, 0.79)	(10.49, 81.91)	(1.68, 9.41)	(-0.78, 0.74)
SwinUNETR [72]	Mean $\pm$ Std	0.54 $\pm$ 0.11	0.67 $\pm$ 0.15	0.48 $\pm$ 0.16	0.55 $\pm$ 0.09	0.64 $\pm$ 0.11	27.88 $\pm$ 18.53	4.67 $\pm$ 2.45	-0.22 $\pm$ 0.39
	Median (IQR)	0.56 (0.15)	0.67 (0.17)	0.51 (0.13)	0.57 (0.17)	0.65 (0.13)	21.50 (19.08)	3.89 (3.35)	-0.24 (0.43)
	(Min, Max)	(0.34, 0.66)	(0.40, 0.93)	(0.23, 0.90)	(0.38, 0.67)	(0.45, 0.78)	(10.68, 84.24)	(2.12, 10.79)	(-0.74, 0.91)
D^2 -RD-UNet (Ours)	Mean $\pm$ Std	0.59 $\pm$ 0.08	0.71 $\pm$ 0.15	0.54 $\pm$ 0.13	0.60 $\pm$ 0.07	0.71 $\pm$ 0.10	23.27 $\pm$ 21.42	3.65 $\pm$ 2.57	-0.17 $\pm$ 0.37
	Median (IQR)	0.60 (0.08)	0.72 (0.17)	0.54 (0.13)	0.60 (0.07)	0.74 (0.09)	15.00 (10.90)	3.00 (1.70)	-0.24 (0.34)
	(Min, Max)	(0.42, 0.72)	(0.40, 0.92)	(0.27, 0.87)	(0.45, 0.74)	(0.43, 0.82)	(8.00, 93.37)	(1.40, 12.92)	(-0.69, 0.93)
D^2 -RD-UNet-VCC (Ours)	Mean $\pm$ Std	0.60 $\pm$ 0.08	0.70 $\pm$ 0.15	0.55 $\pm$ 0.14	0.61 $\pm$ 0.07	0.72 $\pm$ 0.10	22.59 $\pm$ 16.19	3.57 $\pm$ 2.18	-0.15 $\pm$ 0.39
	Median (IQR)	0.62 (0.09)	0.72 (0.17)	0.56 (0.11)	0.61 (0.09)	0.72 (0.09)	15.63 (15.22)	2.90 (2.02)	-0.21 (0.30)
	(Min, Max)	(0.42, 0.72)	(0.43, 0.93)	(0.27, 0.89)	(0.49, 0.74)	(0.37, 0.82)	(8.00, 68.54)	(1.37, 10.82)	(-0.71, 1.02)
D^2 -RD-UNet-VCC (4D) (Ours)	Mean $\pm$ Std	0.61 $\pm$ 0.08	0.71 $\pm$ 0.14	0.57 $\pm$ 0.13	0.62 $\pm$ 0.07	0.74 $\pm$ 0.07	22.17 $\pm$ 14.02	3.28 $\pm$ 1.40	-0.12 $\pm$ 0.41
	Median (IQR)	0.62 (0.07)	0.70 (0.14)	0.54 (0.15)	0.63 (0.07)	0.74 (0.06)	17.66 (17.03)	2.95 (1.84)	-0.31 (0.32)
	(Min, Max)	(0.42; 0.73)	(0.43; 0.90)	(0.27; 0.92)	(0.48; 0.74)	(0.51; 0.82)	(7.68; 64.57)	(1.44; 6.24)	(-0.69; 1.15)
Portal Vessel (PV)	Metric	DSC	PR	RE	MCC	clDice	HD95 [mm]	ASSD [mm]	RVD
nnU-Net [69]	Mean $\pm$ Std	0.63 $\pm$ 0.05	0.66 $\pm$ 0.14	0.64 $\pm$ 0.13	0.64 $\pm$ 0.05	0.64 $\pm$ 0.09	25.34 $\pm$ 19.90	3.99 $\pm$ 2.65	0.03 $\pm$ 0.42
	Median (IQR)	0.62 (0.05)	0.62 (0.26)	0.66 (0.21)	0.64 (0.04)	0.66 (0.09)	17.46 (8.74)	3.14 (1.41)	-0.11 (0.69)
	(Min, Max)	(0.49, 0.72)	(0.46, 0.86)	(0.41, 0.86)	(0.50, 0.72)	(0.37, 0.77)	(10.79, 84.86)	(1.87, 13.72)	(-0.44, 0.87)
VNet [70]	Mean $\pm$ Std	0.61 $\pm$ 0.07	0.66 $\pm$ 0.14	0.60 $\pm$ 0.14	0.62 $\pm$ 0.06	0.60 $\pm$ 0.10	23.59 $\pm$ 18.04	4.15 $\pm$ 2.89	-0.01 $\pm$ 0.42
	Median (IQR)	0.60 (0.07)	0.67 (0.24)	0.62 (0.22)	0.62 (0.07)	0.62 (0.09)	17.36 (12.90)	3.42 (2.07)	-0.08 (0.66)
	(Min, Max)	(0.42, 0.72)	(0.44, 0.86)	(0.32, 0.87)	(0.44, 0.72)	(0.34, 0.80)	(11.05, 90.40)	(1.70, 14.92)	(-0.49, 0.80)
UNet-AG [71]	Mean $\pm$ Std	0.61 $\pm$ 0.07	0.64 $\pm$ 0.14	0.62 $\pm$ 0.15	0.62 $\pm$ 0.07	0.60 $\pm$ 0.09	24.43 $\pm$ 16.22	4.11 $\pm$ 1.96	0.04 $\pm$ 0.42
	Median (IQR)	0.60 (0.07)	0.62 (0.26)	0.64 (0.25)	0.62 (0.07)	0.60 (0.10)	19.43 (7.90)	3.54 (2.17)	-0.10 (0.72)
	(Min, Max)	(0.42, 0.74)	(0.46, 0.85)	(0.37, 0.83)	(0.42, 0.74)	(0.43, 0.79)	(10.82, 67.91)	(1.71, 8.39)	(-0.47, 0.79)
SwinUNETR [72]	Mean $\pm$ Std	0.61 $\pm$ 0.09	0.68 $\pm$ 0.14	0.60 $\pm$ 0.18	0.62 $\pm$ 0.08	0.62 $\pm$ 0.10	24.82 $\pm$ 11.83	4.21 $\pm$ 1.83	-0.04 $\pm$ 0.44
	Median (IQR)	0.62 (0.10)	0.69 (0.22)	0.65 (0.25)	0.64 (0.09)	0.63 (0.10)	20.17 (15.88)	3.81 (3.28)	-0.11 (0.65)
	(Min, Max)	(0.40, 0.73)	(0.47, 0.92)	(0.30, 0.91)	(0.41, 0.73)	(0.43, 0.79)	(10.05, 48.28)	(1.50, 7.13)	(-0.58, 0.91)
D^2 -RD-UNet (Ours)	Mean $\pm$ Std	0.66 $\pm$ 0.05	0.67 $\pm$ 0.14	0.68 $\pm$ 0.13	0.67 $\pm$ 0.05	0.68 $\pm$ 0.07	23.31 $\pm$ 17.27	3.33 $\pm$ 1.83	0.09 $\pm$ 0.40
	Median (IQR)	0.65 (0.06)	0.68 (0.24)	0.67 (0.20)	0.68 (0.05)	0.67 (0.08)	16.49 (12.70)	2.78 (2.24)	0.00 (0.70)
	(Min, Max)	(0.54, 0.75)	(0.47, 0.89)	(0.50, 0.91)	(0.55, 0.75)	(0.49, 0.83)	(9.43, 73.58)	(1.27, 8.85)	(-0.43, 0.81)
D^2 -RD-UNet-VCC (Ours)	Mean $\pm$ Std	0.67 $\pm$ 0.05	0.69 $\pm$ 0.15	0.69 $\pm$ 0.14	0.68 $\pm$ 0.05	0.71 $\pm$ 0.08	22.43 $\pm$ 20.17	3.08 $\pm$ 2.31	0.08 $\pm$ 0.40
	Median (IQR)	0.68 (0.06)	0.69 (0.26)	0.68 (0.22)	0.69 (0.05)	0.71 (0.08)	16.72 (6.19)	2.62 (1.35)	0.03 (0.70)
	(Min, Max)	(0.51, 0.75)	(0.50, 0.93)	(0.49, 0.93)	(0.51, 0.75)	(0.45, 0.83)	(9.27, 97.44)	(1.28, 11.92)	(-0.46, 0.82)
D^2 -RD-UNet-VCC (4D) (Ours)	Mean $\pm$ Std	0.68 $\pm$ 0.05	0.67 $\pm$ 0.14	0.73 $\pm$ 0.11	0.69 $\pm$ 0.04	0.72 $\pm$ 0.05	20.07 $\pm$ 10.47	2.74 $\pm$ 1.02	0.18 $\pm$ 0.41
	Median (IQR)	0.68 (0.09)	0.69 (0.27)	0.72 (0.16)	0.68 (0.06)	0.71 (0.04)	17.20 (16.91)	2.57 (2.07)	0.08 (0.76)
	(Min, Max)	(0.57; 0.75)	(0.46; 0.93)	(0.53; 0.92)	(0.59; 0.75)	(0.62; 0.81)	(8.37; 44.82)	(1.31; 4.38)	(-0.43; 0.91)
Unified Vessel (UV)	Metric	DSC	PR	RE	MCC	clDice	HD95 [mm]	ASSD [mm]	RVD
nnU-Net [69]	Mean $\pm$ Std	0.63 $\pm$ 0.07	0.68 $\pm$ 0.14	0.62 $\pm$ 0.12	0.64 $\pm$ 0.06	0.74 $\pm$ 0.09	18.50 $\pm$ 18.21	2.70 $\pm$ 1.93	-0.02 $\pm$ 0.39
	Median (IQR)	0.63 (0.10)	0.69 (0.18)	0.62 (0.12)	0.64 (0.07)	0.76 (0.06)	12.06 (9.27)	2.24 (1.02)	-0.06 (0.38)
	(Min, Max)	(0.45, 0.72)	(0.43, 0.91)	(0.33, 0.83)	(0.49, 0.73)	(0.43, 0.82)	(7.07, 78.47)	(1.20, 10.32)	(-0.54, 0.78)
VNet [70]	Mean $\pm$ Std	0.61 $\pm$ 0.08	0.73 $\pm$ 0.14	0.56 $\pm$ 0.13	0.63 $\pm$ 0.06	0.73 $\pm$ 0.10	18.56 $\pm$ 19.80	2.88 $\pm$ 2.57	-0.18 $\pm$ 0.35
	Median (IQR)	0.62 (0.08)	0.76 (0.20)	0.56 (0.14)	0.64 (0.08)	0.75 (0.06)	12.22 (5.88)	2.14 (1.02)	-0.20 (0.36)
	(Min, Max)	(0.37, 0.72)	(0.49, 0.94)	(0.25, 0.80)	(0.43, 0.72)	(0.32, 0.83)	(7.00, 75.72)	(1.09, 12.75)	(-0.68, 0.55)
UNet-AG [71]	Mean $\pm$ Std	0.62 $\pm$ 0.06	0.74 $\pm$ 0.14	0.57 $\pm$ 0.13	0.63 $\pm$ 0.05	0.73 $\pm$ 0.07	18.29 $\pm$ 16.77	2.64 $\pm$ 1.35	-0.18 $\pm$ 0.36
	Median (IQR)	0.63 (0.07)	0.76 (0.16)	0.59 (0.16)	0.64 (0.06)	0.73 (0.09)	13.46 (6.52)	2.24 (1.08)	-0.24 (0.38)
	(Min, Max)	(0.50, 0.72)	(0.47, 0.95)	(0.34, 0.79)	(0.55, 0.72)	(0.52, 0.82)	(8.00, 74.43)	(1.37, 6.86)	(-0.64, 0.69)
SwinUNETR [72]	Mean $\pm$ Std	0.61 $\pm$ 0.05	0.73 $\pm$ 0.14	0.57 $\pm$ 0.13	0.63 $\pm$ 0.04	0.73 $\pm$ 0.05	17.84 $\pm$ 15.50	2.54 $\pm$ 0.90	-0.16 $\pm$ 0.37
	Median (IQR)	0.62 (0.05)	0.74 (0.18)	0.58 (0.17)	0.63 (0.05)	0.72 (0.07)	13.56 (6.85)	2.33 (0.91)	-0.22 (0.41)
	(Min, Max)	(0.51, 0.69)	(0.49, 0.94)	(0.37, 0.84)	(0.54, 0.69)	(0.64, 0.83)	(8.77, 77.41)	(1.56, 5.26)	(-0.61, 0.65)
D^2 -RD-UNet (Ours)	Mean $\pm$ Std	0.64 $\pm$ 0.05	0.71 $\pm$ 0.14	0.61 $\pm$ 0.11	0.65 $\pm$ 0.04	0.75 $\pm$ 0.07	16.90 $\pm$ 16.26	2.37 $\pm$ 1.43	-0.08 $\pm$ 0.34
	Median (IQR)	0.64 (0.09)	0.73 (0.20)	0.62 (0.12)	0.64 (0.08)	0.77 (0.07)	11.02 (4.98)	1.92 (0.56)	-0.16 (0.40)
	(Min, Max)	(0.56, 0.72)	(0.46, 0.93)	(0.40, 0.82)	(0.59, 0.72)	(0.52, 0.83)	(6.40, 62.38)	(1.21, 7.68)	(-0.57, 0.80)
D^2 -RD-UNet-VCC (Ours)	Mean $\pm$ Std	0.64 $\pm$ 0.05	0.71 $\pm$ 0.14	0.62 $\pm$ 0.11	0.65 $\pm$ 0.05	0.76 $\pm$ 0.07	16.75 $\pm$ 16.09	2.33 $\pm$ 1.39	-0.07 $\pm$ 0.35
	Median (IQR)	0.65 (0.09)	0.72 (0.20)	0.62 (0.10)	0.64 (0.07)	0.77 (0.07)	10.95 (5.30)	1.92 (0.62)	-0.14 (0.39)
	(Min, Max)	(0.57, 0.72)	(0.46, 0.93)	(0.41, 0.83)	(0.59, 0.73)	(0.54, 0.83)	(6.32, 62.38)	(1.16, 7.41)	(-0.56, 0.82)
D^2 -RD-UNet-VCC (4D) (Ours)	Mean $\pm$ Std	0.66 $\pm$ 0.05	0.70 $\pm$ 0.13	0.65 $\pm$ 0.10	0.66 $\pm$ 0.05	0.77 $\pm$ 0.04	15.08 $\pm$ 12.01	2.10 $\pm$ 0.74	-0.02 $\pm$ 0.35
	Median (IQR)	0.66 (0.07)	0.72 (0.18)	0.66 (0.09)	0.66 (0.08)	0.77 (0.04)	10.63 (6.05)	1.88 (0.62)	-0.09 (0.40)
	(Min, Max)	(0.56; 0.72)	(0.44; 0.91)	(0.43; 0.88)	(0.58; 0.74)	(0.67; 0.82)	(6.00; 57.08)	(0.96; 3.55)	(-0.48; 0.98)

transformer-based approaches, their advantage over convolutional architectures in medical image segmentation tasks remained limited, with CNNs keep showing superior performance in many scenarios.

In summary, these results suggest that the VCC algorithm improved the automatic segmentation results in both HV and PV, with particular improvement in PV, preserving topology and continuity and reducing false positives and false negatives. Overall, the D^2 -RD-UNet framework, particularly in its 4D and VCC configurations, demonstrated robust adaptability and superior performance across all three segmentation tasks. The VCC algorithm increased RE while maintaining PR, particularly in the HV and PV tasks. These findings underscore the effectiveness of D^2 -RD-UNet in addressing the challenges of vascular segmentation, such as boundary errors and false positives, even when using less precise public GT masks. Incorporating vesselness filters in the 4D configuration further enhanced the detection of fine and branching vascular structures, reducing errors and preserving topological integrity.

5.3.2. Qualitative results

The qualitative evaluation of the 3D vascular reconstructions from the 20 subjects in the 3D-IRCADb-01 dataset highlighted the strengths of the D^2 -D2-RD-UNet variants, especially those enhanced by VCC and vesselness filter 4D approaches, compared to nnU-Net, VNet, UNet-AG. Similar to the observations made with the AIMS-HPV-385 TeS dataset, the D^2 -D2-RD-UNet models have shown superior continuity and preservation of fine vascular branches, maintaining topological coherence even in complex bifurcations. In the case depicted in Fig. 9(1), the models minimized disconnections and gaps that were prevalent in SOTA models, achieving higher DSC and cDice values while capturing the finest details often missed by VNet and nnU-Net. The VCC-enhanced D^2 -D2-RD-UNet achieved the highest performance (DSC = 0.74, cDice = 0.78), emphasizing its ability to generalize well the vascular reconstructions. In the pathological case (Fig. 9(2)), the GT annotations lacked sufficient detail, which was particularly noteworthy. Consequently, the D^2 -D2-RD-UNet variants trained on a robust and enriched dataset identified a significantly greater number of structures compared to the GT, leading to a slight reduction in metrics such as DSC and cDice despite achieving visually and anatomically superior reconstructions. This behavior was particularly evident in pathological regions, where irregular geometries and vessel deformations challenged competing models like nnU-Net and VNet. Although SwinUNETR did not qualitatively deviate significantly from the performance of CNN-based models, on the 3D-IRCADb-01 dataset it appeared to generalize even better in physiological scenarios, outperforming traditional CNN-based architectures in capturing anatomical detail and continuity. However, its generalization ability dropped in pathological cases, where the performance became less consistent. In contrast, our D^2 -D2-RD-UNet variants maintained higher robustness across different anatomical conditions, particularly in preserving topological accuracy and vascular continuity. These observations showed us that D^2 -D2-RD-UNet variants reliably reconstructed detailed blood vessel pathways, with vasculature filters that improved detection of small branches. Their consistency across datasets and ability to recover details often missed due to imperfect ground truth labels made them suitable for surgical planning and clinical imaging.

5.4. Local evaluation metrics

In this section, we present the results of our radius-based local metric analysis, conducted using the 4D model of our approach, which integrated both 4D vesselness information and VCC. Given the global performance of this model, as demonstrated in Sections 5.2 and 5.3, D^2 -RD-UNet-VCC (4D) was selected as the basis for this radius-based local evaluation. The analysis provided insights into the model's behavior in pathological and non-pathological scenarios. We evaluated performance across two datasets: AIMS-HPV-385 TeS and 3D-IRCADb-01.

5.4.1. Radius-based approach on AIMS-HPV-385 TeS

Using the radius-based metric for the AIMS-HPV-385 TeS dataset, we observed a clear dependence of segmentation accuracy on branch size. Smaller-radius branches, particularly those below 1 mm, consistently showed lower DSC values, reflecting the model difficulty in capturing finer vascular structures. This behavior is evident in the HV histogram of Fig. 10(left), where the peak frequency of branches with radii below 1 mm is equivalent to a drop of the DSC values in the range of 0.00–0.30 according to the color bar. Similarly, in the PV histogram of Fig. 10(right), branches with radii below 1 mm showed DSC values predominantly between 0.00–0.50, further underscoring the challenges in segmenting finer structures. Similar observations can be made from Figure E.2 in the Appendix, which presents the histograms for the full set of metrics: DSC, PR, RE, MCC, and HD95. Fig. 11 illustrates this behavior for a specific pathological subject, highlighting errors near the tumor mass, where structural distortions worsened segmentation challenges. In Fig. 11a, false negative branches, present in the GT but absent in the predictions, are more visible near distal regions and smaller branches, particularly near the lesion. Conversely, Fig. 11b shows false positive branches, predicted by the model but absent in the GT, sporadically distributed along larger branches or near bifurcations. These errors are associated with localized DSC values as low as 0.00–0.20, highlighting regions where the model struggles with topological consistency. Overall, the radius-based branch local metric provided valuable insights that reflects localized challenges that may be missed by global metrics, e.g., DSC. Quantitatively, larger branches with radii above 3 mm were segmented more accurately, with DSC values approaching 0.80–1.00 in both HV and PV. These results indicated that the model performed well on larger structures, while its performance degraded significantly for smaller branches. The HV histogram also reveals a high frequency of branches within the radius range of 0.60–1.00 mm, which coincides with a pronounced decrease in global DSC values, emphasizing the importance of addressing errors in this range.

5.4.2. Radius-based approach on 3D-IRCADb-01

Using the radius-based metric for the 3D-IRCADb-01 dataset, we observed trends in segmentation accuracy that shared some similarities with the AIMS-HPV-385 TeS dataset but also exhibited notable differences, reflecting the challenges posed by the lower quality and variability of the 3D-IRCADb-01 GT annotations. Smaller-radius branches, particularly those below 1 mm, showed consistently lower DSC values, similar to AIMS-HPV-385. This trend is visible in the HV histogram of Fig. 12(left), where branches with radii below 1 mm correspond to DSC values primarily in the range of 0.00–0.30. Similarly, in the PV histogram of Fig. 12(right), branches with radii below 1 mm exhibit DSC values between 0.00–0.50, underscoring the difficulty of segmenting finer structures. Similar observations can be made from Figure E.3 in the Appendix, which presents the histograms for the full set of metrics: DSC, PR, RE, MCC, and HD95. However, unlike AIMS-HPV-385, the 3D-IRCADb-01 histograms made it harder to determine a clear line of separation between smaller-radius branches and larger ones concerning segmentation performances. In AIMS-HPV-385, the transition from lower DSC values (below 1 mm) to higher accuracy (above 3 mm) is more distinct. In 3D-IRCADb-01, intermediate-radius branches (1–3 mm) show a broader, less consistent range of DSC values. For instance, in both HV and PV, branches in this range frequently achieved DSC values between 0.30–0.70, but their distribution was irregular, suggesting that the model's performance varied more unpredictably for this dataset. This variability may stem from the dataset's smaller size (20 subjects versus 58 in AIMS-HPV-385) and the inconsistent quality of the GT annotations. Fig. 13 provides additional insights into segmentation errors for a specific pathological subject (IR-15). False negatives branches (present in GT but missing in predictions) are concentrated in distal regions and smaller branches near the lesion,

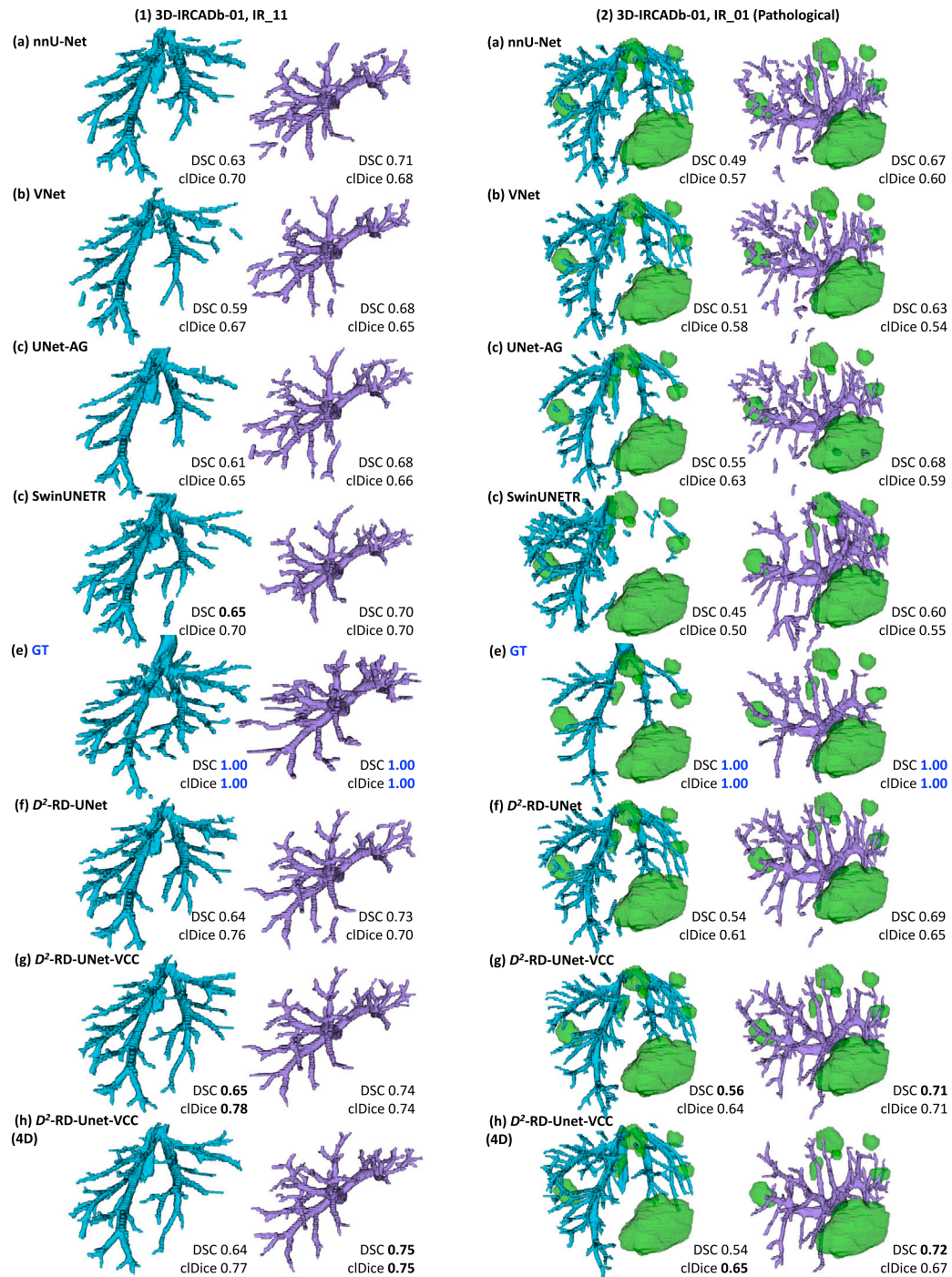


Fig. 9. 3D-IRCADb-01 qualitative comparison of vessel segmentation methods. The figure shows HV (cyan) and PV (purple) segmentations for two subjects: (1) Subject IR_11 and (2) Subject IR_01, a pathological case with the tumor label publicly provided. Methods compared include SOTA models (nnU-Net, VNet, U-Net-AG, SwinUNETR) and our three D^2 -RD-UNet variants. (e) shows the GT segmentations for reference. DSC and cDice are reported for each method, with the best values highlighted in bold.

a behavior consistent with AIMS-HPV-385. However, in 3D-IRCADb-01, false negatives regions also appeared in large-radius structures like the cava vein, partially under-segmented and marked in red. These errors, visible in Fig. 13a, suggest potential undersegmentation of the model's predictions or discrepancies in the GT annotations for such large structures. False positives branches (predicted by the model but absent in GT), as shown in Fig. 13b, remain sporadically distributed along larger branches or bifurcations, with localized DSC values dropping to as low as 0.00–0.20, similar AIMS-HPV-385. Quantitatively, larger-radius branches (above 3 mm) generally achieve DSC values

close to 0.80–1.00, as observed in both datasets. However, the less distinct performance separation between branch sizes in 3D-IRCADb-01 highlighted the limitations of this dataset. Large structures, such as the cava vein, occasionally exhibit low DSC values, likely due to GT inconsistencies or partial under-segmentation. These observations suggested that while the model generalizes reasonably well to 3D-IRCADb-01, the variability in performance, especially for intermediate-radius branches, reflected the challenges of working with datasets of lower annotation quality. In summary, the 3D-IRCADb-01 dataset showed patterns similar to AIMS-HPV-385 for small-radius branches, but the lack of a clear

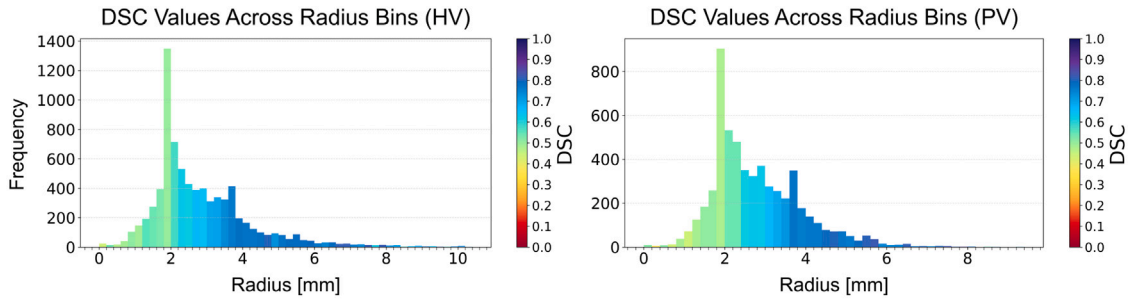


Fig. 10. Histogram representing the frequency of branches radius across the whole AIMS-HPV-385 TeS dataset of 58 subjects. Each bar, with a bin size of 0.2 mm, is color-coded based on the color bar, indicating the DSC value for the corresponding interval.

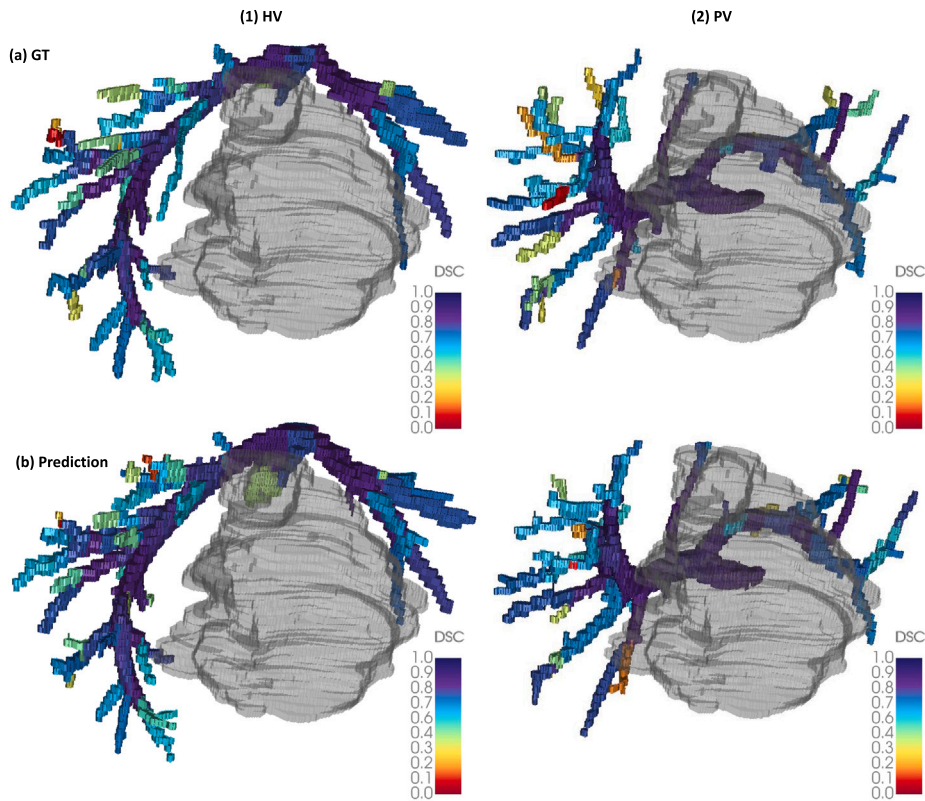


Fig. 11. Visualization of the DSC metric based on the color bar for a pathological subject from AIMS-HPV-385 TeS. (a) maps of DSC values onto the GT branches, with red segments indicating branches present in GT but absent in predictions. (b) maps of DSC values onto the predictions, with red segments indicating branches present in predictions but absent in GT. The tumor mass is highlighted in gray.

separation in performance between smaller and larger branches made it more difficult to draw definitive conclusions about the model's behavior. The additional segmentation challenges for large structures, such as the vena cava, further emphasized the need for careful evaluation when using 3D-IRCADb-01 as a benchmark dataset. This analysis highlighted the importance of both high-quality annotations and diverse datasets to achieve a robust and clinically reliable segmentation evaluation.

5.5. Efficiency analysis

In this study, we assessed the efficiency of our proposed 3D and 4D models compared to SOTA architectures. The evaluation was based on a standardized set of metrics, including total training time (hours), inference time per batch (seconds), trainable parameter count (millions), computational complexity (FLOPs), and GPU memory usage (GB). To ensure a fair comparison, all experiments were conducted on

an NVIDIA A100 GPU with 40 GB memory, using a consistent input size of $192 \times 192 \times 64$ for 3D networks and $192 \times 192 \times 64 \times 5$ for 4D networks. A batch size of 4 was used across all models. Table 8 shows the performance metrics for the SOTA models and for the D^2 -RD-UNet in its 3D and 4D variants. The VCC variant was excluded from the table, as its inclusion did not impact the training procedure or efficiency metrics, focusing the comparison only on 3D and 4D architectures.

The results highlighted clear trade-offs between computational cost and model performance. Among the 3D models, D^2 -RD-UNet stood out with a strong balance between efficiency and performance. It achieved competitive FLOPs (5355.48 GFLOPs), a relatively low memory footprint (2.97 GB), and a moderate training time (4.58 h), making it an efficient option for applications with constrained resources. Similarly, nnU-Net and VNet demonstrated high efficiency, with VNet achieving the fastest inference time (1.26 s per volume) and the smallest memory usage (3.32 GB). These attributes made 3D models valuable when

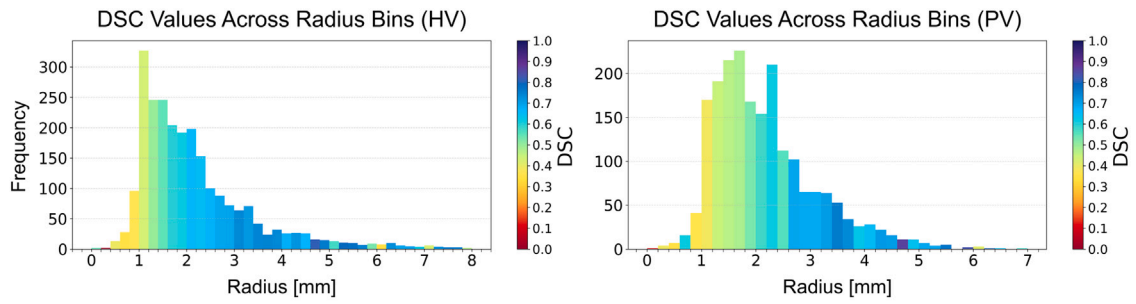


Fig. 12. Histogram representing the frequency of branches radius across the whole 3D-IRCADb-01 dataset of 20 subjects. Each bar, with a bin size of 0.2 mm, is color-coded based on the color bar, indicating the DSC value for the corresponding interval. For larger diameters, the DSC values fluctuate because major vessels, such as the cava and portal vein at the hepatic hilum, often extend beyond the segmentation limits imposed by the annotation protocol of the AIMS-HPV-385 dataset.

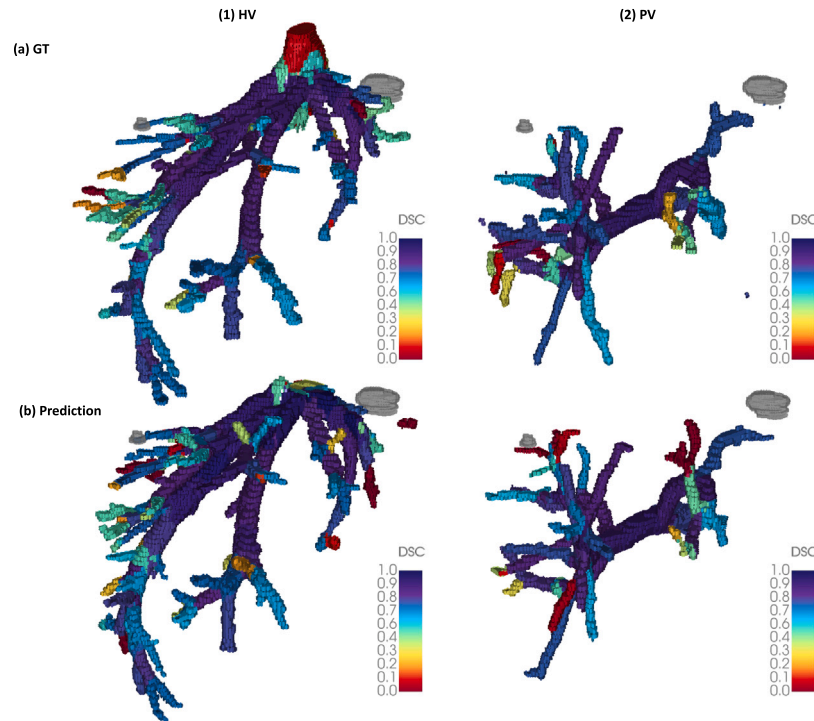


Fig. 13. Visualization of the DSC metric based on the color bar for pathological subject IR_15 from 3D-IRCADb-01. (a) maps of DSC values onto the GT branches, with red segments indicating branches present in GT but absent in predictions. (b) maps of DSC values onto the predictions, with red segments indicating branches present in predictions but absent in GT. The tumor mass is highlighted in gray.

Table 8

Computational performance of different trained models. The table compares SOTA models (nnU-Net, VNet, UNet-AG, SwinUNETR) and the D^2 -RD-UNet variants regarding the computational performance of their configurations. The VCC algorithm is not included, as it does not affect these metrics. Results provide an overview of the computational requirements and efficiency of each model.

Model	Total training time (h)	Inference time (s)	Parameters (M)	FLOPs (GFLOPs)	Memory (GB)
nnU-Net	2.17	6.39	9.39	2736.30	3.45
VNet	2.10	1.26	10.09	2781.67	3.32
UNet-AG	6.87	2.50	11.50	2900.00	3.40
SwinUNETR	24.76	7.80	40.02	6200.00	5.46
D^2 -RD-UNet	4.58	4.18	7.24	5355.48	2.97
D^2 -RD-UNet (4D)	19.29	6.90	7.32	5571.17	3.63

computational resources are limited or real-time inference is required, except for SwinUNETR. It required the largest computational budget among 3D models, with 40.02M parameters, 6200 GFLOPs, 5.46 GB memory usage, and a training time of 24.76 h. In contrast, the D^2 -RD-UNet (4D) demonstrated the best performance in previous results, with a superior ability to predict the complex structures of blood vessels effectively. However, these performance gains came at a significant

computational cost, so carefully considering the trade-off between accuracy and resource demands for specific applications is essential. For instance, the D^2 -RD-UNet (4D) required 5571.17 GFLOPs and 3.63 GB of memory, with a notably longer training time of 19.29 h. While the 4D architectures offered enhanced modeling capacity and potential gains in segmentation accuracy, the increased computational demands

must be carefully weighed against the practical constraints of available hardware and processing time.

Generating the 4D input required a preprocessing step to compute a multi-channel hypervolume using multiple vesselness filters. On average, on a three-sample batch from 3D-IRCADb-01 images, preprocessing took 72 s (Frangi), 67 s (Sato), 231 s (OOF), and 1776 s (RORPO). While filters like Frangi and Sato add manageable overhead, methods such as RORPO introduced substantial delays that could hinder clinical deployment, particularly in time-sensitive workflows. It is further important to note that the processing time is mostly related to image size.

Overall, this analysis emphasized the need to consider both performance gains and computational costs when choosing between 3D and 4D models. While 4D models can provide richer contextual representations, their significantly higher training and inference costs than 3D models may not always justify the incremental improvements, particularly in resource-constrained settings. Although tested on an A100 (40 GB), all models had a memory footprint under 6 GB, suggesting feasibility on more common clinical-grade GPUs such as the RTX 2080 Ti (11 GB) or even mid-range cards like the RTX 3060 (12 GB). Future research should focus on optimizing 4D architectures to reduce their computational burden while preserving their ability to leverage accurate results.

6. Conclusions

In this work, we introduced the D^2 -RD-UNet, a dual-stage, dual-class segmentation framework specifically designed for the accurate and reliable segmentation of hepatic and portal veins. The architecture proved its ability to address the complex challenges of vascular segmentation, including class imbalance, anatomical variability, and the preservation of vascular topology. By leveraging a robust preprocessing pipeline, vessel-specific enhancements, and novel postprocessing corrections, our approach achieved superior performance compared to SOTA models across multiple datasets and evaluation metrics. The 4D variant of the D^2 -RD-UNet, exploiting vesselness filters, improved segmentation quality by capturing finer vascular details and maintaining topological continuity. The integration of the VCC postprocessing algorithm further contributed to these results by enhancing the model's ability to preserve the continuity and integrity of vascular structures, addressing a critical challenge in vascular segmentation. While the 4D variant demonstrated superior performance, this came with higher computational costs. In contrast, including VCC did not significantly impact computational efficiency, making it a practical and effective addition to the framework. Overall, the D^2 -RD-based models presented in this work showcased improved performances over SOTA models, with a clear qualitative and quantitative edge on both the AIMS-HPV-385 and 3D-IRCADb-01 test sets. The combination of advanced architectural features and tailored pre- and postprocessing techniques allowed these models to consistently outperform existing approaches, delivering more accurate and reliable segmentations across different datasets. Introducing a radius-based evaluation metric further underscored the clinical relevance of our method, enabling precise assessments of segmentation quality across different vascular sizes. Moreover, this metric may also serve as a confidence index for the model, identifying regions where predictions may be less reliable and warrant additional radiological attention. This feature can be notably valuable in HPB surgery, where accurate vascular segmentation is crucial for preoperative planning and surgical decision-making. However, it is also important to note that not all HPB procedures require detailed segmentation of fine-grained vessels. Therefore, while our current model shows strong capabilities in capturing finer vascular details, further advancements in this area will be a focus for future developments, ensuring tailored solutions for scenarios requiring detailed capillary-level segmentation. Future work will also focus on optimizing the computational efficiency of the 4D variant, reducing training costs, and exploring its applicability to

other anatomically complex vascular systems. Extending the framework to incorporate multimodal imaging data, such as MRI, could improve its robustness and generalizability. The D^2 -RD-UNet framework, enriched by VCC and evaluated with the radius-based metric, represents an advancement in liver vessel segmentation, potentially enhancing preoperative planning and improving clinical outcomes in liver surgeries.

Abbreviations and symbols

See Table 9.

CRediT authorship contribution statement

Matteo Cavicchioli: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Andrea Moglia:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Guillaume Garret:** Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization. **Martina Puglia:** Writing – review & editing, Data curation. **Antoine Vacavant:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Giacomo Pugliese:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Data curation, Conceptualization. **Pietro Cerveri:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Ethics statement

This research do not involve humans or animals. Therefore no institutional review board and patient informed consent were required.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors did not use any tool or software based on generative AI and AI-assisted technologies.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Andrea Moglia reports financial support was provided by Polytechnic of Milan Department of Electronics Information and Bioengineering. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.combiomed.2025.110530>.

Data availability

Data are available upon request.

Table 9
List of abbreviations and mathematical symbols used throughout the manuscript and Appendix.

(a) Abbreviations		(b) Mathematical Symbols	
Abbr.	Definition	Symbol	Definition
ACM	Active Contour Model	α, β, C	Frangi filter parameters
AG	Attention Gate	α_1, α_2	Sato filter parameters
AP	Arterial Phase	B	Batch size
ASSD	Average Symmetric Surface Distance	C	Number of output classes
BN	Batch Normalization	Cnl_{HV}	Hepatic vein centerlines
clDice	Centerline Dice	Cnl_{PV}	Portal vein centerlines
CNN	Convolutional Neural Network	D	Depth
CT	Computed Tomography	e_i	Eigenvectors of Hessian
CTA	Computed Tomography Angiography	e_{ki}^{HV}	i th endpoint of k th HV centerline
D^2 -RD-UNet	Dual-Stage Dual-Class Residual-Dense UNet	e_{lj}^{PV}	j th endpoint of l th PV centerline
DL	Deep Learning	End_k^{HV}	Endpoints of k th HV centerline
DSC	Dice Similarity Coefficient	End_l^{PV}	Endpoints of l th PV centerline
EDT	Euclidean Distance Transform	F_{Frangi}	Frangi vesselness
FLOPs	Floating Point Operations Per Second	F_{OOF}	OOF vesselness
GT	Ground Truth	F_{Sato}	Sato vesselness
GPU	Graphics Processing Unit	FN	False Negatives
HD95	95th percentile Hausdorff Distance	FP	False Positives
HPB	Hepato-Biliary-Pancreatic	H	Height
HU	Hounsfield Unit	$Hess$	Hessian matrix
HV	Hepatic Vein	I	Image intensity
IQR	Interquartile Range	$I \oplus SE$	Morphological dilation
IVC	Inferior Vena Cava	$I \ominus SE$	Morphological erosion
LV	Liver Volume	$I \circ SE$	Morphological opening
MCC	Matthews Correlation Coefficient	$I \bullet SE$	Morphological closing
MRI	Magnetic Resonance Imaging	λ_i	Hessian eigenvalues
MSD	Medical Segmentation Decathlon	λ_c	$\min(-\lambda_2, -\lambda_3)$
nnU-Net	No-new-U-Net	N	Number of voxels ($W \times H \times D$)
NCCN®	National Comprehensive Cancer Network®	$npix_c$	Voxels in class c
NN	Nearest Neighbor	$Npix$	Total dataset voxels
OOF	Optimally Oriented Flux	p_k, q_l	Number of HV, PV endpoints
PR	Precision	Q	Oriented flux matrix
PV	Portal Vein	R	Spherical kernel radius
PVP	Portal Venous Phase	R_a	Line-like vs. planar object ratio
RE	Recall	R_b	Blob-like vs. planar object ratio
ROI	Region of Interest	S	Frobenius norm of Hessian
RORPO	Ranking Orientation Responses of Path Operators	SE	Structuring element
RVD	Relative Volume Difference	σ	Gaussian std. dev.
SOTA	State-of-the-Art	$\sigma_{min}, \sigma_{max}$	Min and max scales
STD	Standard Deviation	T_{prec}	Topology precision
SwinUNETR	Swin Transformer UNet	TP	True Positives
TeS	Test Set	TN	True Negatives
TrS	Training Set	$Tsens$	Topology sensitivity
UV	Unified Vessel (HV + PV)	w_c	Class weight
VCC	Vessel Connectivity Correction	W	Width
		X	Input tensor
		γ	Focal loss parameter

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